PROVEIT RESEARCH CONTINUATION

Bilinear Tail Identities, Polylogarithmic Resonances, and Stieltjes Boundary Products

Complete moment reconstruction at orders ten and twelve

11 October 2026

Abstract

We continue the exact-identity programme in the current ProveIt manuscript and its five incoming research packages. First we classify all rational certificates formed from the two adjacent products of independently colored tails. Such a certificate exists precisely when the quotient of the denominator polynomials is a constant times $h(z)h(z+1)$ with h rational. This gives two closed independently shifted Nielsen-beta kernels, a rank-two identity, all mixed polygamma derivatives, a Gamma– Hurwitz rectangular primitive, and a Hermitian polylogarithm-tail law. Second, a holomorphic combination of two neighboring polylogarithms evaluates an entire family of Mellin resonances through finite factored shift operators; its mixed jets have explicit zeta and Stieltjes coefficients. Third, a completed two-variable Hurwitz product kernel evaluates every finite-part product having one undifferentiated Stieltjes factor and one positive argument derivative, including an exact integration-by-parts endpoint law. Finally, we complete the optimal moment-prefix conjecture for every unequal even pair whose smaller order is ten or twelve. The proofs combine exact source reductions, one five-by-five certificate, an effective analytic threshold, and a fully replayed finite-field certificate. All convergence conventions and remaining conjectures are stated explicitly. Historical priority beyond the inspected literature and repository corpus is not asserted.

Repository snapshot. `da2792fc9cc8ef19c6a314be6c65e1437fbdc66b`

Deliverables. Complete source and PDF, exact verification programs, recorded certificates, numerical diagnostics, and integration notes.

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1 Research targets, source status, and conventions

The relevant source state is the ProveIt commit `da2792fc9cc8ef19c6a314be6c65e1437fbdc66b`, dated 11 October 2026, 21:44:07 UTC. We inspected the unified manuscript and the five archives then present in `docs/incoming`: the alternating-tail, finite-difference-determinant, higher-quadratic-jet, higher-spectral-defect, and quartic-reconstruction reports. Their mathematical status must be read together. In particular, a question left open in one report can already have a partial answer in another report written from an earlier snapshot. The exact paths and selected source passages are recorded in `provenance/sources.json`.

The new results concern identities, finite reductions, and exact reconstruction. The only estimates used below justify interchange of limits or close a determinant argument; they are not the research target.

The proofs share a concrete ordering principle: enforce cancellation before summing or specializing parameters. The tail certificates remove the linear tail terms. The Mellin operators annihilate finite Lerch corrections. The Hurwitz completions remove the moving coordinate resonances. The centered moment transform isolates finitely many zeta coordinates. Keeping the parameters independent until these steps are complete preserves the mixed derivatives and endpoint information.

1.1 What advances and what is inherited

Source target	Contribution in this article	Scope
Two adjacent bilinear tail certificates [2]	Necessary and sufficient rational-factorization criterion; exact cutoff identity; independent-shift evaluations	Complete classification within the stated two-product certificate class.
Independently shifted tail identities	Two Nielsen-beta kernels, all mixed argument derivatives, and a Gamma–Hurwitz rectangular primitive	Two freely varying shifts for adjacent quadratic denominators; arbitrary independent tail orders are not claimed.
Polylogarithm derivatives and antiderivatives	A holomorphic neighboring-order combination, finite Mellin-resonance operators, and mixed Stieltjes jets	All complex polylogarithm orders on explicitly stated domains.
Undifferentiated Stieltjes edge [4]	Complete joint-pole subtraction and finite ordinary-zeta algorithm	Every product with exactly one undifferentiated factor; an endpoint law holds for all derivative orders.
Optimal first-q moment prefix [2, 3]	Complete smaller-order-ten and smaller-order-twelve families	All even $q > 10$ and $q > 12$, respectively; formal coordinate rank.

The bilinear classification and the two completed reconstruction families answer explicit project questions. The Stieltjes completion answers a previously excluded boundary case. The Mellin operator and kernel formulas are proved here and were not located in the inspected project corpus; their classical ingredients are identified when used. The literature search is targeted, not an

exhaustive priority determination. Products of zeta tails, Hurwitz product integrals, rational summation, and Mellin transforms all have substantial prior literature [5, 6, 7, 9].

1.2 Notation and analytic conventions

The Hurwitz zeta function is initially

$$\zeta(s,a)=\sum_{n=0}^{\infty}(n+a)^{-s},\qquad \Re s>1,\quad \Re a>0,$$

and is then continued. Its generalized Stieltjes constants satisfy

$$\zeta(1+u,a)=\frac{1}{u}+\sum_{m=0}^{\infty}\frac{(-1)^m}{m!}\gamma_m(a)u^m. \tag{1.1}$$

Thus $\gamma_0(a)=-\psi(a)$ and $\gamma_m=\gamma_m(1)$. A prime on $\zeta(s,a)$ differentiates the order s; a superscript $\gamma_m^{(r)}(a)$ differentiates the argument a. We write

$$H_n^{(r)}=\sum_{k=1}^{n}k^{-r},\qquad H_0^{(r)}=0,\qquad \psi^{(r)}(a)=(-1)^{r+1}r!\zeta(r+1,a)\quad (r\ge 1).$$

Bernoulli numbers use $B_1=-1/2$ and $t/(e^t-1)=\sum_{j\ge 0}B_jt^j/j!$. These normalizations agree with the standard sources [10, 11].

All powers and logarithms use the principal branch on the initial slit or right-half-plane domains. The logarithm of Γ on the right half-plane is the holomorphic branch real on the positive axis. An ordinary sum is the limit of the displayed partial sums. A finite part is always specified relative to a coordinate, and a spectral continuation is kept distinct from an ordinary divergent sum. Local normal convergence means locally uniform absolute convergence.

The phrase *formal coordinate rank* refers to rational linear algebra in the explicitly defined source coordinate space. It does not assert independence of the evaluated periods. The finite-field computations are exact arithmetic certificates with complete replay; the analytic arguments remain written mathematical proofs. No proof-assistant formalization is claimed.

2 Classification of rational bilinear tail certificates

The latest alternating-tail report asks whether allowing both adjacent products gives a useful classification for two different denominator polynomials. We answer that question exactly. The classification is relative to an explicit certificate class; failure of its criterion does not exclude longer telescopers or other special-function identities.

2.1 The certificate class and its factorization criterion

Let $P,Q\in\mathbb{C}[z]$ be nonzero, let $\alpha,\beta,t\neq 0$, and suppose two sequences satisfy

$$E_{n-1}=E_n+\frac{\alpha^n}{P(n)},\qquad F_{n-1}=F_n+\frac{\beta^n}{Q(n)}. \tag{2.1}$$

For rational functions A,B, consider

$$T_n=t^n\{A(n+1)E_nF_{n+1}+B(n+1)E_{n+1}F_n\}. \tag{2.2}$$

We ask that, after using (2.1), the coefficients of E_n and F_n in $T_n - T_{n-1}$ both vanish identically. The remaining term is then a rational weight times E_nF_n. This is an algebraic condition on the certificate, independent of any conjectured numerical relation among the tails.

Theorem 2.1 (Complete two-adjacent-product classification)**.** *A nonzero rational certificate* (2.2) *has the stated cancellation property if and only if*

$$t^2\alpha\beta = 1, \qquad \frac{P(z)}{Q(z)} = \rho h(z)h(z+1) \tag{2.3}$$

for a nonzero constant ρ and a nonzero rational function h. In that case all certificates for the chosen t are scalar multiples of

$$A(z) = Q(z)h(z), \qquad B(z) = \kappa Q(z)h(z+1), \qquad \kappa = -t\beta. \tag{2.4}$$

Writing $C = A + B$, their exact difference is

$$T_n - T_{n-1} = t^n\{C(n+1) - t^{-1}C(n)\}E_nF_n. \tag{2.5}$$

Thus, when the rational factorization exists, there are exactly two outer-color choices t, and the certificate space for either choice is one-dimensional.

Here "nonzero" means that (A, B) is not the zero pair. At a special parameter value the resulting weight can vanish identically, or the evaluated tails can make the certificate zero. Such degeneracies do not change the formal cancellation classification. An example is the equal-shift limit of the alternating kernel below, whose nontrivial diagonal identity is recovered by holomorphic continuation.

Proof. Direct expansion gives the quadratic coefficient $A(n+1) + B(n+1) - t^{-1}\{A(n) + B(n)\}$. The linear coefficients vanish exactly when

$$\beta\frac{A(z+1)}{Q(z+1)} + t^{-1}\frac{B(z)}{Q(z)} = 0, \qquad \alpha\frac{B(z+1)}{P(z+1)} + t^{-1}\frac{A(z)}{P(z)} = 0. \tag{2.6}$$

If one of A, B is zero, both are zero. Otherwise set $h = A/Q$ and $R = P/Q$. The first equation gives $B = -t\beta Qh(z+1)$; substitution into the second gives

$$\frac{R(z+1)}{R(z)} = t^2\alpha\beta\frac{h(z+2)}{h(z)}. \tag{2.7}$$

A nonzero rational function r satisfies $r(z+c)/r(z) \to 1$ as $z \to \infty$ for every fixed c. Therefore $t^2\alpha\beta = 1$. Equation (2.7) now says that $R(z)/\{h(z)h(z+1)\}$ is a rational function with period one. A nonconstant rational periodic function would have infinitely many poles, or, if polynomial, could not have a nonzero period. It is therefore the constant ρ. This proves necessity.

Conversely, (2.3) and (2.4) verify (2.6) by substitution, proving the difference identity. If h_1, h_2 are two solutions, their quotient r satisfies $r(z)r(z+1) =$ constant and hence $r(z+2) = r(z)$. It is constant by the same rational-periodicity argument. This proves the uniqueness assertion. □

2.2 An effective criterion on the divisor of the rational function

The condition in (2.3) does not require an infinite search for h. Reduce $R = P/Q$ first, and group its finite zeros and poles into equivalence classes modulo integer translation. For an orbit $\xi + \mathbb{Z}$, put

$$r_j = \operatorname{ord}_{z=\xi+j} R(z).$$

Only finitely many r_j are nonzero; poles have negative multiplicity.

Proposition 2.2 (Alternating divisor test)**.** *The factorization $R(z) = \rho h(z)h(z+1)$ with rational h exists if and only if, for every such orbit,*

$$\sum_{j\in\mathbb{Z}} (-1)^j r_j = 0. \tag{2.8}$$

When this holds, the multiplicity of h at $\xi + j$ is

$$h_j = \sum_{k\geq 0} (-1)^k r_{j+k}. \tag{2.9}$$

The constant ρ then matches the leading coefficients at infinity.

Proof. The multiplicities must satisfy $r_j = h_j + h_{j+1}$. The unique solution vanishing above the support of r is (2.9). It also vanishes below that support exactly when (2.8) holds. This constructs a rational h with the required finite divisor. The quotient $R/(hh(\cdot+1))$ has neither finite zeros nor poles, and its order at infinity is also zero, because $\sum r_j = 2\sum h_j$. It is therefore a nonzero constant. Changing the orbit representative only multiplies (2.8) by a sign. □

This is a finite algebraic test on the reduced rational function. In particular, its degree at infinity must be even. Generic quadratic polynomials with four zeros in distinct integer-translation orbits fail the test. By contrast,

$$\frac{(z+a)(z+a+1)}{(z+b)(z+b+1)} = \frac{z+a}{z+b}\frac{z+a+1}{z+b+1} \tag{2.10}$$

passes for arbitrary independent a, b. This example gives the two-shift identities in the next section. The divisor argument is an elementary instance of rational difference-equation reasoning; the contribution here is its exact use to classify this bilinear tail certificate and all compatible colors.

2.3 The finite endpoint and the infinite identity

Assume now that $\deg P = d \geq 2$, $\deg Q = e \geq 2$, and neither polynomial vanishes at a positive integer. Let

$$0 < |\alpha|, |\beta| \leq 1, \qquad \alpha \neq 1, \quad \beta \neq 1,$$

and define the ordinary tails

$$E_n = \sum_{k>n} \frac{\alpha^k}{P(k)}, \qquad F_n = \sum_{k>n} \frac{\beta^k}{Q(k)}. \tag{2.11}$$

Suppose the equivalent conditions of Theorem 2.1 hold. A zero or pole of h at a positive integer would force, by (2.3), alternating nonzero multiplicities at all later integers, which is impossible for a rational function. Consequently $h(1)$ is finite and nonzero. The rational functions A, B are regular at nonnegative integers after removal of any removable singularities.

Theorem 2.3 (Bilinear summation with its exact upper endpoint)**.** *Put* $W(n) = C(n+1) - t^{-1}C(n)$ *and*

$$\mathcal{E} = \beta h(1)E_0 - \frac{t^{-1}}{\rho h(1)}F_0 - t^{-1}C(0)E_0F_0. \tag{2.12}$$

For every integer $N \geq 0$,

$$\sum_{n=0}^{N} t^n W(n)E_nF_n = \mathcal{E} + T_N. \tag{2.13}$$

Moreover,

$$\boxed{\sum_{n=0}^{\infty} t^n W(n)E_nF_n = \mathcal{E},} \tag{2.14}$$

and the series converges absolutely. For fixed P, Q, h*, it converges locally normally in the admissible colors away from* 1*, with either locally chosen branch of* t*.*

Proof. Sum (2.5) for $1 \leq n \leq N$. Since $E_1 = E_0 - \alpha/P(1)$ and $F_1 = F_0 - \beta/Q(1)$,

$$T_0 = C(1)E_0F_0 - \beta h(1)E_0 - \frac{\alpha\kappa}{\rho h(1)}F_0.$$

Use $\alpha\kappa = -t^{-1}$ and restore the $n = 0$ summand. This gives (2.13), including the lower-endpoint term $C(0)E_0F_0$.

Summation by parts, using bounded partial geometric sums, gives $E_n = O(|\alpha|^{n+1}n^{-d})$ and $F_n = O(|\beta|^{n+1}n^{-e})$. The bounds are locally uniform as stated. The factorization forces $\deg h = (d-e)/2$, so A, B, C are $O(n^{(d+e)/2})$. Since $|t| = |\alpha\beta|^{-1/2}$, the summand and T_N are respectively bounded by a constant times

$$|\alpha\beta|^{n/2}n^{-(d+e)/2}, \qquad |\alpha\beta|^{N/2}N^{-(d+e)/2}.$$

The first is summable and the second tends to zero. This proves the infinite identity and the local normal convergence assertion. □

The finite identity remains true outside these sufficient convergence hypotheses whenever the recurrences make sense. Any additional ordinary or regularized limit needs its own endpoint analysis.

2.4 A Hermitian polylogarithm-tail law

The two inner colors need not agree. Taking conjugate colors gives an Hermitian identity with explicit dependence on the color. Its weights are positive for the power-polynomial specialization below.

Corollary 2.4 (Conjugate-color completion)**.** *Let* P *be a real polynomial of degree at least two with no positive integer zero. For* $0 < \theta < 2\pi$*, write*

$$E_n(\theta) = \sum_{k>n} \frac{e^{ik\theta}}{P(k)}.$$

Then

$$\sum_{n\geq 0}\{P(n+1) - P(n)\}|E_n(\theta)|^2 = \cot\frac{\theta}{2}\,\Im E_0(\theta) - \Re E_0(\theta) - P(0)|E_0(\theta)|^2. \tag{2.15}$$

For every integer $d \geq 2$, in particular,

$$\boxed{\sum_{n\geq 0}\{(n+1)^d - n^d\}\left|\mathrm{Li}_d(e^{i\theta}) - \sum_{k=1}^{n}\frac{e^{ik\theta}}{k^d}\right|^2 = \cot\frac{\theta}{2}\,\Im\,\mathrm{Li}_d(e^{i\theta}) - \Re\,\mathrm{Li}_d(e^{i\theta}).} \tag{2.16}$$

Proof. In Theorem 2.3, take $Q = P$, $h = 1$, $\rho = 1$, $\alpha = e^{i\theta}$, $\beta = e^{-i\theta}$, and $t = 1$. Then $C = (1 - e^{-i\theta})P$ and $F_n = \overline{E_n}$. Divide by $1 - e^{-i\theta}$ and use

$$\frac{e^{-i\theta}E - \overline{E}}{1 - e^{-i\theta}} = -\Re E + \cot(\theta/2)\Im E.$$

□

At the Gaussian color and order two, this gives the explicit identity

$$\boxed{\sum_{n\geq 0}(2n+1)\left|\mathrm{Li}_2(i) - \sum_{k=1}^{n}\frac{i^k}{k^2}\right|^2 = G + \frac{\pi^2}{48},} \tag{2.17}$$

where $G = \sum_{k\geq 0}(-1)^k/(2k+1)^2$ is Catalan's constant. Here $\mathrm{Li}_2(i) = -\pi^2/48 + iG$, obtained by separating even and odd terms of the defining series. At $\theta = \pi$ the formula recovers the ordinary-outer alternating square law of [2]. For $d > 2$, its limit as $\theta \to 0$ is also an ordinary identity:

$$\sum_{n\geq 0}\{(n+1)^d - n^d\}\{\zeta(d) - H_n^{(d)}\}^2 = 2\zeta(d-1) - \zeta(d).$$

Indeed the absolute tail is bounded by $O(n^{1-d})$, giving a summable majorant $O(n^{1-d})$ after multiplication by the weight. The right side follows by differentiating the color at zero, justified by $d > 2$. This limit is not asserted at $d = 2$, where the ordinary series diverges. The new classification supplies these formulas as corollaries; they are not separate claims of global literature priority.

3 Independent shifts, polygamma kernels, and Gamma primitives

The adjacent-quadratic factorization (2.10) makes both permitted outer colors explicit. The ordinary-outer branch has a kernel of rank at most two. The alternating-outer branch has a divided-difference kernel. Both admit all mixed argument derivatives.

3.1 Two complete Nielsen-beta kernels

For $\Re z > 0$, define Nielsen's beta function and its adjacent difference by

$$\mathcal{B}(z) = \sum_{j\geq 0}\frac{(-1)^j}{j+z} = \frac{1}{2}\left\{\psi\left(\frac{z+1}{2}\right) - \psi\left(\frac{z}{2}\right)\right\}, \tag{3.1}$$

$$U(z) = 2\mathcal{B}(z) - \frac{1}{z} = \sum_{j\geq 0}\frac{(-1)^j}{(j+z)(j+z+1)} = \int_0^1 x^{z-1}\frac{1-x}{1+x}\,dx. \tag{3.2}$$

The beta series converges by summation by parts; the series defining U converges absolutely. Its last representation follows from $\mathcal{B}(z) + \mathcal{B}(z+1) = 1/z$. These are standard descriptions of Nielsen's function [8]; the identities below concern products of its independently shifted adjacent differences.

Theorem 3.1 (Ordinary and alternating outer kernels)**.** *For $\Re z, \Re w > 0$, the following series are locally normally convergent:*

$$K(z,w) = \sum_{n\geq 0} (2n+z+w)U(n+z)U(n+w), \tag{3.3}$$

$$S(z,w) = \sum_{n\geq 0} (-1)^n U(n+z)U(n+w). \tag{3.4}$$

They have the exact evaluations

$$\boxed{K(z,w) = \frac{z}{2w}U(z) + \frac{w}{2z}U(w) - \left(zw - \frac{z+w}{2}\right)U(z)U(w),} \tag{3.5}$$

and, for $z \neq w$,

$$\boxed{S(z,w) = \frac{1}{2}U(z)U(w) + \frac{z^2U(z) - w^2U(w)}{2zw(z-w)}.} \tag{3.6}$$

The apparent singularity of (3.6) on $z = w$ is removable:

$$S(z,z) = \frac{1}{2}U(z)^2 + \frac{U(z)}{z} + \frac{1}{2}U'(z). \tag{3.7}$$

Proof. Take

$$P(k) = (k+z-1)(k+z), \qquad Q(k) = (k+w-1)(k+w), \qquad h(k) = \frac{k+z-1}{k+w-1}, \quad \rho = 1,$$

and $\alpha = \beta = -1$ in Theorem 2.3. Then $E_n = (-1)^{n+1}U(n+z)$ and $F_n = (-1)^{n+1}U(n+w)$. For $t = 1$, the certificate coefficients are

$$A(k) = (k+z-1)(k+w), \qquad B(k) = (k+z)(k+w-1).$$

Hence

$$C(k) = 2k^2 + 2(z+w-1)k + 2zw - z - w, \qquad C(k+1) - C(k) = 2(2k+z+w).$$

The lower endpoint (2.12), divided by two, gives (3.5). For $t = -1$, change the sign of B; now $C(k) = z - w$ and $C(k+1) + C(k) = 2(z-w)$. The endpoint gives (3.6). Its diagonal limit is (3.7).

One can also read the exact truncation error from the proof. For the ordinary kernel it is

$$\begin{aligned}\sum_{n=0}^{N} (2n+z+w)U(n+z)U(n+w) = K(z,w) \\ - \frac{1}{2}\{(N+z)(N+w+1)U(N+z)U(N+w+1) \\ + (N+z+1)(N+w)U(N+z+1)U(N+w)\}.\end{aligned} \tag{3.8}$$

Finally, (3.2) gives $U(n+z) = O(n^{-2})$ locally uniformly in the right half-plane. The ordinary and alternating summands are therefore $O(n^{-3})$ and $O(n^{-4})$, respectively. This also justifies the holomorphic diagonal limit. □

For $z = w$, formula (3.5) becomes

$$K(z,z) = U(z) - z(z-1)U(z)^2,$$

the adjacent-quadratic square law in [2]. The off-diagonal formula contains two independent shifts; it is not obtained by setting two parameters inside a single tail and then squaring that tail.

3.2 Explicit harmonic and half-shift evaluations

Let $L = \log 2$. The elementary values

$$U(1) = 2L - 1, \qquad U(2) = \frac{3}{2} - 2L, \qquad U(1/2) = \pi - 2$$

give

$$\boxed{\sum_{n\geq 0}(2n+3)U(n+1)U(n+2) = 2(1-L)^2,} \tag{3.9}$$

$$\boxed{\sum_{n\geq 0}(4n+3)U(n+1/2)U(n+1) = (\pi+2)L - 2,} \tag{3.10}$$

$$\boxed{\sum_{n\geq 0}(-1)^n U(n+1)^2 = 2L^2 - \frac{\pi^2}{12}.} \tag{3.11}$$

For the last identity use $U'(1) = 1 - \pi^2/6$ in (3.7). These are ordinary convergent series. They can be written entirely with harmonic numbers: if

$$H_n^{\mathrm{alt}} = \sum_{k=1}^{n} \frac{(-1)^{k-1}}{k}, \qquad u_n = 2(-1)^n\{L - H_n^{\mathrm{alt}}\} - \frac{1}{n+1},$$

then $u_n = U(n+1)$, so (3.9) is $\sum_{n\geq 0}(2n+3)u_n u_{n+1} = 2(1-L)^2$, while (3.11) evaluates $\sum_{n\geq 0}(-1)^n u_n^2$. The cancellation in the displayed expression for u_n is part of the formula; individual unsummed constant pieces are not to be separated.

3.3 Rank two and all mixed polygamma derivatives

Put $f(z) = zU(z)$. Formula (3.5) has the simpler equivalent form

$$K(z,w) = f(z)\mathcal{B}(w) + \mathcal{B}(z)f(w) - f(z)f(w). \tag{3.12}$$

Consequently, for any three right-half-plane points z_1, z_2, z_3,

$$\det\left[\sum_{n\geq 0}(2n + z_i + z_j)U(n+z_i)U(n+z_j)\right]_{1\leq i,j\leq 3} = 0. \tag{3.13}$$

More precisely the two-point determinant is

$$K(z,z)K(w,w) - K(z,w)^2 = -\{f(z)\mathcal{B}(w) - \mathcal{B}(z)f(w)\}^2. \tag{3.14}$$

These are identities in the complex variables. They follow by writing the matrix as the product of its two coordinate columns, the fixed matrix $\left(\begin{smallmatrix}-1 & 1\\ 1 & 0\end{smallmatrix}\right)$, and the transpose of the coordinate columns. They concern this parameter-dependent weighted kernel, not the unweighted Gram matrix.

Corollary 3.2 (Every pair of argument derivatives)**.** *For all integers $r, s \geq 0$ and $\Re z, \Re w > 0$,*

$$\begin{aligned}\sum_{n\geq 0}\{&(2n+z+w)U^{(r)}(n+z)U^{(s)}(n+w) + rU^{(r-1)}(n+z)U^{(s)}(n+w)\\ &+ sU^{(r)}(n+z)U^{(s-1)}(n+w)\}\\ &= f^{(r)}(z)\mathcal{B}^{(s)}(w) + \mathcal{B}^{(r)}(z)f^{(s)}(w) - f^{(r)}(z)f^{(s)}(w).\end{aligned} \tag{3.15}$$

A term with zero prefactor is omitted. The corresponding alternating identity is obtained by applying $\partial_z^r\partial_w^s$ to (3.6), with its holomorphic diagonal interpretation.

Proof. Differentiating the integral (3.2) gives

$$U^{(r)}(n+z) = \int_0^1 x^{n+z-1} \log^r x \frac{1-x}{1+x}\, dx = O(n^{-r-2})$$

locally uniformly. This proves the required normal convergence after every fixed number of derivatives. Apply the product rule to (3.12). □

Every term is explicitly a polygamma expression, because

$$U^{(r)}(z) = 2^{-r}\left\{\psi^{(r)}\left(\frac{z+1}{2}\right) - \psi^{(r)}\left(\frac{z}{2}\right)\right\} + \frac{(-1)^{r+1}r!}{z^{r+1}}. \tag{3.16}$$

For example, $r = s = 1$, $z = w = 1$ yields

$$\sum_{n\geq 0}\{(2n+2)U'(n+1)^2 + 2U(n+1)U'(n+1)\} = \frac{\pi^2 \log 2}{3} - 4\log^2 2. \tag{3.17}$$

The two lower-derivative terms in (3.15) are required by the derivative of the weight. Omitting them would give a different, generally false identity.

3.4 An explicitly anchored Gamma–Hurwitz primitive

The rank-two form makes a two-variable antiderivative finite as well. Define the following holomorphic functions on $\Re z > 0$:

$$\mathcal{L}(z) = \log\Gamma\left(\frac{z+1}{2}\right) - \log\Gamma\left(\frac{z}{2}\right), \tag{3.18}$$

$$\mathcal{Q}(z) = 2z\mathcal{L}(z) - 4\left\{\zeta'\left(-1, \frac{z+1}{2}\right) - \zeta'\left(-1, \frac{z}{2}\right)\right\}, \tag{3.19}$$

$$\mathcal{U}(z) = 2\mathcal{L}(z) - \log z. \tag{3.20}$$

They satisfy

$$\mathcal{L}'(z) = \mathcal{B}(z), \qquad \mathcal{Q}'(z) = zU(z), \qquad \mathcal{U}'(z) = U(z). \tag{3.21}$$

For the middle identity, differentiate the Hurwitz shift law:

$$\partial_a \zeta'(-1,a) = -\zeta(0,a) + \zeta'(0,a) = a - \frac{1}{2} + \log\Gamma(a) - \frac{1}{2}\log(2\pi).$$

It follows that the derivative of the brace in (3.19) is $1/4 + \mathcal{L}(z)/2$, giving (3.21). Only standard Hurwitz special values are used here [10].

Fix anchors z_0, w_0 in the right half-plane. For any holomorphic V, use the notation

$$\Delta_z V(n) = V(n+z) - V(n+z_0), \qquad \Delta_w V(n) = V(n+w) - V(n+w_0).$$

Theorem 3.3 (Rectangular Gamma–Hurwitz summation identity)**.** *For z, z_0, w, w_0 in the right half-plane,*

$$\begin{aligned}&\sum_{n\geq 0}\{\Delta_z\mathcal{Q}(n)\Delta_w\mathcal{U}(n) + \Delta_z\mathcal{U}(n)\Delta_w\mathcal{Q}(n)\}\\ &\quad = \Delta_z\mathcal{Q}(0)\Delta_w\mathcal{L}(0) + \Delta_z\mathcal{L}(0)\Delta_w\mathcal{Q}(0) - \Delta_z\mathcal{Q}(0)\Delta_w\mathcal{Q}(0).\end{aligned} \tag{3.22}$$

The series is ordinarily and locally normally convergent, and its value vanishes on either anchor face. Thus (3.22) has no unspecified constants of integration. If $\mathcal{R}$ denotes its displayed right side, its exact finite version is

$$\sum_{n=0}^{N}\{\Delta_z\mathcal{Q}(n)\Delta_w\mathcal{U}(n)+\Delta_z\mathcal{U}(n)\Delta_w\mathcal{Q}(n)\}$$
$$=\mathcal{R}-\frac{1}{2}\{\Delta_z\mathcal{Q}(N)\Delta_w\mathcal{Q}(N+1)+\Delta_z\mathcal{Q}(N+1)\Delta_w\mathcal{Q}(N)\}. \tag{3.23}$$

Proof. Integrate (3.12) in z from z_0 to z and in w from w_0 to w. In the defining summand,

$$(2n+z+w)U(n+z)U(n+w)=f(n+z)U(n+w)+U(n+z)f(n+w).$$

Equations (3.21) give the left side of (3.22); the rank-two expression gives its right side. The domain is simply connected, so the chosen paths do not affect the result. Uniformly on compact parameter and anchor sets, $\Delta_z\mathcal{Q}(n)=O(n^{-1})$ and $\Delta_z\mathcal{U}(n)=O(n^{-2})$. The products are $O(n^{-3})$, which justifies the integrations and proves the convergence assertion. Integrating the two separated products in the exact upper endpoint (3.8) gives (3.23). □

At equal anchors and equal endpoints, the formula reduces to the one-parameter identity

$$\sum_{n\geq 0}\Delta_z\mathcal{Q}(n)\Delta_z\mathcal{U}(n)=\Delta_z\mathcal{Q}(0)\Delta_z\mathcal{L}(0)-\frac{1}{2}\{\Delta_z\mathcal{Q}(0)\}^2.$$

This connects an ordinary series of Gamma-ratio and Hurwitz-zeta antiderivatives directly to its lower endpoint. The explicit anchor also makes every further argument derivative unambiguous.

4 A Mellin transform with exact cancellation at every integer exponent

The next family keeps the polylogarithm order complex. This is essential: an identity proved only at positive integer orders would not justify the order derivatives that produce Stieltjes constants below. The starting ingredients are the classical Fermi–Dirac integral for the polylogarithm, the Lerch integral, and Euler's beta integral [9, 12, 13]. The contribution developed here is a finite factored operator for all rational kernels in the family, together with joint continuation through the Mellin and polylogarithm singularities. The full-line family and its operator were not located in the inspected manuscript or five newest incoming reports. This is a statement about the inspected corpus, not a claim of worldwide priority. Related work on integrals of products of polylogarithms includes Freitas and Sofo [14, 15]; their quoted finite-interval problems should not be confused with the full-line transform proved here.

Let

$$\Omega=\mathbb{C}\setminus(-\infty,0],\qquad(\mathsf{E}f)(s)=f(s-1),\qquad\mathscr{Z}(s)=s\zeta(s+1).$$

The function $\mathscr{Z}$ is entire, with $\mathscr{Z}(0)=1$. Products such as $(\mathsf{E})_k$ are ordinary polynomials in the shift operator, using the rising-factorial convention; in particular $(\mathsf{E})_2=\mathsf{E}(\mathsf{E}+1)$. For $m\geq 1$ define

$$\mathcal{I}_m(t,s;\lambda)=\int_0^\infty\frac{x^{t-1}\operatorname{Li}_s(-\lambda x)}{(1+x)^m}\,dx,\qquad -1<\Re t<m,\quad s\in\mathbb{C},\quad\lambda\in\Omega. \tag{4.1}$$

Here $x^{t-1} = \exp((t-1)\log x)$ uses the real logarithm, while the polylogarithm uses its principal argument branch. Thus its argument never crosses $[1,\infty)$ for the displayed values of λ. The stated t strip is a common convergence domain for every order. Special orders, notably negative integers, can have larger convergence strips; no maximal-strip assertion for each individual order is intended.

4.1 Analyticity of the ordinary integrals

Lemma 4.1. *The integral (4.1) is absolutely convergent and jointly holomorphic in its three variables on the displayed domain. Every fixed number of derivatives with respect to s, t, and λ may be taken under the integral sign. A t derivative inserts a factor $\log x$, and $\lambda\partial_\lambda$ replaces the order s by $s-1$.*

Proof. We record a compact-set bound that also covers negative and noninteger orders. If s and λ range over compact subsets of $\mathbb{C}$ and Ω, respectively, then

$$\mathrm{Li}_s(-\lambda x) = O(x) \quad (x \downarrow 0), \qquad \mathrm{Li}_s(-\lambda x) = O((1+\log x)^B) \quad (x \geq 1)$$

for a sufficiently large fixed B. The first bound follows from the normally convergent power series near zero. For the second, choose an integer N with $\Re(s+N) > 0$ throughout a slightly larger compact set and use

$$\mathrm{Li}_{s+N}(-\lambda x) = -\frac{1}{\Gamma(s+N)} \int_0^\infty u^{s+N-1} \frac{\lambda x}{e^u + \lambda x}\, du.$$

Apply $(x\partial_x)^N$. For a complex variable on rays uniformly separated from the negative real axis, every fixed logarithmic derivative of $w/(1+w)$ is bounded by $C\min(|w|,1)$. Splitting the u integral at $\log x + C_0$ therefore gives the asserted logarithmic bound. The identity $(x\partial_x)^N \mathrm{Li}_{s+N} = \mathrm{Li}_s$ holds first by the power series and then by analytic continuation. Cauchy's formula on the slightly larger compact parameter sets supplies the same sort of bounds for all fixed parameter derivatives.

The resulting majorants at the two endpoints are $x^{\Re t}$ and $x^{\Re t - m - 1}(1+\log x)^B$. They are locally uniformly integrable throughout the common strip used in (4.1). Inserting any fixed power of $\log x$ preserves integrability. The standard dominated holomorphy theorem completes the proof. □

4.2 The paired Lerch expression

For $\Re a > 0$, first on $|\lambda| < 1$ within Ω, put

$$\mathcal{H}_a(s,\lambda) = \mathrm{Li}_s(\lambda) - \lambda^a \Phi(\lambda, s, a), \qquad \lambda^a = \exp(a\,\mathrm{Log}\,\lambda). \tag{4.2}$$

Its two terms must be continued together. This convention has a concrete ordinary-integral explanation.

Lemma 4.2. *The function (4.2) extends jointly holomorphically to $s \in \mathbb{C}$, $\Re a > 0$, and $\lambda \in \Omega$. For $\Re s > 0$,*

$$\mathcal{H}_a(s,\lambda) = \frac{1}{\Gamma(s)} \int_0^\infty u^{s-1} \frac{w - w^a}{1-w}\, du, \qquad w = \lambda e^{-u}, \quad w^a = \exp(a(\mathrm{Log}\,\lambda - u)). \tag{4.3}$$

The apparent singularity at $w = 1$ has the removable value $a-1$. Consequently

$$\mathcal{H}_a(s,1) = \zeta(s) - \zeta(s,a), \tag{4.4}$$

$$\mathcal{H}_M(s,\lambda) = \sum_{n=1}^{M-1} \frac{\lambda^n}{n^s} \quad (M = 1,2,\ldots). \tag{4.5}$$

The pole of each zeta function in (4.4) cancels.

Proof. For $|\lambda| < 1$ and $\Re s > 0$, the defining geometric series and the gamma integral give (4.3). The quotient is analytic at $w = 1$ because its numerator and denominator vanish to first order and their ratio tends to $a - 1$. It and all fixed u derivatives decay exponentially as $u \to \infty$, locally uniformly in a and λ. They are bounded near $u = 0$, including when $\lambda = 1$.

To obtain the continuation in order without appealing to a divergent Mellin integral, let $K_a(u, \lambda) = (w - w^a)/(1 - w)$. For any integer $N \geq 0$, integration by parts first on $\Re s > 0$ gives

$$\mathcal{H}_a(s, \lambda) = \frac{1}{\Gamma(s + N)} \int_0^\infty u^{s+N-1} (-\partial_u)^N K_a(u, \lambda)\, du.$$

The right side is holomorphic for $\Re s > -N$. All boundary terms vanish in the initial domain; the formulas agree on overlaps by continuation. Since N is arbitrary, they establish entire dependence on s and the claimed joint holomorphy. Substitution of $\lambda = 1$ proves (4.4) first where the separate zeta integrals converge and then everywhere. For a positive integer M, the kernel is the finite geometric sum $w + \cdots + w^{M-1}$, proving (4.5). □

4.3 The full transform and the cancellation polynomial

Define the explicit polynomial

$$P_{m,t}(X) = (1 - t - X)_{m-1} = \sum_{j=0}^{m-1} c_{m,j}(t) X^j. \tag{4.6}$$

Theorem 4.3 (Full Mellin transform)**.** *For every parameter in* (4.1)*,*

$$\boxed{\mathcal{I}_m(t, s; \lambda) = \frac{\pi}{(m-1)! \sin \pi t} \sum_{j=0}^{m-1} c_{m,j}(t) \mathcal{H}_{m-t}(s - j, \lambda).} \tag{4.7}$$

At $t = 0, 1, \ldots, m - 1$ the numerator vanishes identically as a function of s and λ; the apparent poles are removable. At $\lambda = 1$ this becomes the entire-order Hurwitz formula

$$\mathcal{I}_m(t, s; 1) = \frac{\pi}{(m-1)! \sin \pi t} \sum_{j=0}^{m-1} c_{m,j}(t) (\zeta(s - j) - \zeta(s - j, m - t)). \tag{4.8}$$

Proof. First take $m = 1$, $0 < \lambda < 1$, $\Re s > 0$, and $-1 < \Re t < 0$. Insert the Fermi–Dirac integral in Lemma 4.1. Absolute convergence permits interchanging the two integrations. The elementary partial fraction integral

$$\int_0^\infty \frac{x^t}{(1 + x)(x + c)}\, dx = -\frac{\pi}{\sin \pi t} \frac{1 - c^t}{c - 1}, \qquad c > 1,$$

follows from the beta integral. With $c = e^u/\lambda$, it gives

$$\mathcal{I}_1(t, s; \lambda) = \frac{\pi}{\sin \pi t} (\mathrm{Li}_s(\lambda) - \lambda^{1-t} \Phi(\lambda, s, 1 - t)).$$

The preceding two lemmas continue this identity in s and λ, and across the strip $-1 < \Re t < 1$ after removing $t = 0$.

For $m \geq 2$, integrate the derivative of $x^t(1+x)^{1-m}\operatorname{Li}_s(-\lambda x)$ over the positive axis. On the common strip $-1 < \Re t < m-1$ its boundary terms vanish. Using $\mathcal{I}_m(t+1,s;\lambda) = \mathcal{I}_{m-1}(t,s;\lambda) - \mathcal{I}_m(t,s;\lambda)$ yields

$$\mathcal{I}_m(t,s;\lambda) = \frac{m-1-t-\mathsf{E}}{m-1}\,\mathcal{I}_{m-1}(t,s;\lambda). \tag{4.9}$$

Iteration gives $P_{m,t}(\mathsf{E})\mathcal{I}_1/(m-1)!$.

Initially on $|\lambda| < 1$, the elementary Lerch shift relation gives

$$\mathcal{H}_{1-t}(s,\lambda) - \mathcal{H}_{m-t}(s,\lambda) = -\sum_{n=0}^{m-2} \frac{\lambda^{n+1-t}}{(n+1-t)^s}.$$

For every summand, E acts as multiplication by $n+1-t$; $P_{m,t}(n+1-t) = (-n)_{m-1} = 0$. Applying the polynomial therefore annihilates the entire finite shift. This proves (4.7) on a nonempty open strip, with the important positive-real-part argument $m-t$.

At $t = k \in \{0,\dots,m-1\}$, equation (4.5) says that its numerator equals

$$\sum_{n=1}^{m-k-1} \frac{\lambda^n}{n^s}(1-k-n)_{m-1} = 0.$$

Every product in the last sum has a zero factor. Thus all apparent poles inside $-1 < \Re t < m$ are removable. The two sides are holomorphic on that connected strip and agree on an open subset, proving the full domain. Equation (4.4) gives the specialization (4.8). □

For example, the base specialization is

$$\int_0^\infty \frac{x^{t-1}\operatorname{Li}_s(-x)}{1+x}\,dx = \frac{\pi}{\sin \pi t}\big(\zeta(s) - \zeta(s,1-t)\big), \qquad -1 < \Re t < 1. \tag{4.10}$$

The difference on the right is the generalized harmonic function $H_{-t}^{(s)}$. Its singularities at special orders have cancelled before order differentiation is performed.

5 The branch cancellation and a factored formula at resonance

For $|\lambda| < 1$ in Ω, define

$$\mathcal{A}_s(\lambda) = \operatorname{Log}\lambda\,\operatorname{Li}_s(\lambda) - s\operatorname{Li}_{s+1}(\lambda). \tag{5.1}$$

Theorem 5.1 (Continuation through the positive polylogarithm cut)**.** *The paired expression* (5.1) *extends jointly holomorphically to $s \in \mathbb{C}$, $\lambda \in \Omega$. It satisfies*

$$\mathcal{A}_s(\lambda) = \int_0^\infty \frac{\operatorname{Li}_s(-\lambda x)}{x(1+x)}\,dx, \tag{5.2}$$

$$\mathcal{A}_s(1) = -\mathscr{Z}(s), \tag{5.3}$$

$$\lambda\partial_\lambda \mathcal{A}_s(\lambda) = \mathcal{A}_{s-1}(\lambda). \tag{5.4}$$

On the real interval $\lambda > 1$, either consistent boundary value of the two principal polylogarithms in (5.1) *gives the same answer. At $\lambda = 1$ the full expression is removable even at integer orders where its separate gamma and zeta expansions have poles.*

If $u = \operatorname{Log}\lambda$ and $|u| < 2\pi$, then

$$\boxed{\mathcal{A}_s(e^u) = -\sum_{n=0}^{\infty} \mathscr{Z}(s-n)\frac{u^n}{n!}.} \tag{5.5}$$

The series converges normally for s in compact subsets of $\mathbb{C}$ and $|u| < 2\pi$. On the full u disk it describes the continuation of the germ at zero. For the original principal composition one retains $|\Im u| < \pi$, as is automatic when $u = \operatorname{Log}\lambda$.

Proof. Take the removable limit $t \to 0$ in the $m = 1$ transform, initially for $|\lambda| < 1$. The Lerch derivative $\partial_a \Phi(\lambda, s, a) = -s\Phi(\lambda, s+1, a)$ gives

$$\partial_t(\operatorname{Li}_s(\lambda) - \lambda^{1-t}\Phi(\lambda, s, 1-t))\Big|_{t=0} = \operatorname{Log}\lambda\ \operatorname{Li}_s(\lambda) - s\operatorname{Li}_{s+1}(\lambda).$$

This proves (5.2); the integral supplies the claimed joint holomorphic continuation. In particular, it proves that the two boundary values across the positive cut agree, without assigning separate divergent values. For $\Re s > 0$, taking $\lambda \to 1$ from below in (5.1) gives $-s\zeta(s+1)$. Both sides are entire in s, proving (5.3) at every order. Differentiation of (5.2) proves (5.4).

There is also an explicit check of the branch cancellation. For noninteger s, the classical local polylogarithm expansion [9, Eq. 25.12.12] is

$$\operatorname{Li}_s(e^u) = \Gamma(1-s)(-u)^{s-1} + \sum_{n\geq 0} \zeta(s-n)\frac{u^n}{n!}, \qquad |u| < 2\pi,$$

on an initial compatible branch. The two fractional-power terms in $u\operatorname{Li}_s(e^u) - s\operatorname{Li}_{s+1}(e^u)$ cancel because $\Gamma(1-s) = -s\Gamma(-s)$. Combining the remaining series gives precisely (5.5).

Every coefficient $\mathscr{Z}(s-n)$ is entire in s. The zeta functional equation and the gamma-ratio estimate imply, uniformly on a compact set of orders,

$$\frac{|\mathscr{Z}(s-n)|}{n!} \leq C_K n^{B_K}(2\pi)^{-n}$$

for all sufficiently large n. Consequently the series is normally convergent on the claimed product domain. Continuation in s now includes all integer orders and proves the claimed cancellation there. The restriction on the original principal logarithm is stated explicitly to avoid confusing a germ in u with a globally single-valued expression under the exponential map. □

Theorem 5.2 (Every integer Mellin resonance)**.** *For $m \geq 1$, $0 \leq k \leq m-1$, $s \in \mathbb{C}$, and $\lambda \in \Omega$,*

$$\boxed{\int_0^\infty \frac{x^{k-1}\operatorname{Li}_s(-\lambda x)}{(1+x)^m}\,dx = \frac{(\mathsf{E})_k(1-\mathsf{E})_{m-k-1}}{(m-1)!}\mathcal{A}_s(\lambda).} \tag{5.6}$$

In particular,

$$\int_0^\infty \frac{x^{k-1}\operatorname{Li}_s(-x)}{(1+x)^m}\,dx = -\frac{(\mathsf{E})_k(1-\mathsf{E})_{m-k-1}}{(m-1)!}\mathscr{Z}(s). \tag{5.7}$$

The right side is a finite expression in entire functions. Thus every fixed number of order derivatives is valid, including at the values where a displayed individual zeta factor has a pole.

Proof. At $t = 0$, recurrence (4.9) and (5.2) give $\mathcal{I}_m(0, s; \lambda) = (1 - \mathsf{E})_{m-1}\mathcal{A}_s(\lambda)/(m-1)!$. For $k \geq 1$, multiplication of the kernel by $x = (1 + x) - 1$ gives

$$\mathcal{I}_m(k, s; \lambda) = \mathcal{I}_{m-1}(k-1, s; \lambda) - \mathcal{I}_m(k-1, s; \lambda).$$

Induct on k. The claimed polynomials satisfy exactly this recurrence:

$$\begin{aligned}\frac{(X)_{k-1}(1-X)_{m-k-1}}{(m-2)!} &- \frac{(X)_{k-1}(1-X)_{m-k}}{(m-1)!} \\ &= \frac{(X)_{k-1}(1-X)_{m-k-1}}{(m-1)!}((m-1) - (m-k-X)) \\ &= \frac{(X)_k(1-X)_{m-k-1}}{(m-1)!}.\end{aligned}$$

This proves (5.6). Equation (5.3) proves (5.7). □

The first few kernels illustrate the factored operator:

(m, k)	$(m-1)!^{-1}(X)_k(1-X)_{m-k-1}$
$(1, 0)$	1
$(2, 0)$	$1 - X$
$(2, 1)$	X
$(3, 1)$	$(X - X^2)/2$
$(3, 2)$	$(X + X^2)/2$
$(4, 2)$	$(X - X^3)/6$

For instance, uniformly as an entire identity in s,

$$\int_0^\infty \frac{\mathrm{Li}_s(-x)}{(1+x)^2}\,dx = -(s-1)\zeta(s), \tag{5.8}$$

$$\int_0^\infty \frac{x\,\mathrm{Li}_s(-x)}{(1+x)^4}\,dx = \frac{(s-3)\zeta(s-2) - (s-1)\zeta(s)}{6}. \tag{5.9}$$

In (5.8), the value at $s = 1$ is -1, not zero; in (5.9), the value of $(s-3)\zeta(s-2)$ at $s = 3$ is 1. These are ordinary removable values of entire products.

5.1 A bilinear identity isolating a single Stieltjes constant

The same transform has a particularly short bilinear consequence. One polylogarithm has a fixed negative integer order and is therefore a rational function; the other order remains an arbitrary complex variable. No relation is differentiated outside its proven domain.

Corollary 5.3. *For every integer $d \geq 1$, every $s \in \mathbb{C}$, and every $\lambda \in \Omega$,*

$$\boxed{\int_0^\infty \mathrm{Li}_s(-\lambda x)\,\mathrm{Li}_{-d}(-x)\,\frac{dx}{x} = (-1)^d\mathcal{A}_{s-d}(\lambda).} \tag{5.10}$$

The integral is ordinary and absolutely convergent, and is entire in s. At $\lambda = 1$ it equals $(-1)^{d+1}(s-d)\zeta(s-d+1)$, with the removable value understood. Consequently, for every $r \geq 1$,

$$\boxed{\int_0^\infty \mathrm{Li}_{-d}(-x)\,\partial_s^r\,\mathrm{Li}_s(-x)|_{s=d}\,\frac{dx}{x} = (-1)^{d+r}r\gamma_{r-1}.} \tag{5.11}$$

At derivative order zero the integral is $(-1)^{d+1}$.

Proof. Put $D = x\partial_x$ and $f(x) = 1/(1+x)$. Since $f = 1 + \mathrm{Li}_0(-x)$, one has $D^d f = \mathrm{Li}_{-d}(-x)$ for $d \geq 1$. Every $D^j f$ is $O(x^{-1})$ at infinity. At zero it is bounded when $j = 0$ and $O(x)$ when $j \geq 1$. Lemma 4.1 therefore justifies d successive integrations by parts with respect to dx/x; every boundary term vanishes. They give

$$\int_0^\infty \mathrm{Li}_s(-\lambda x) D^d f(x)\,\frac{dx}{x} = (-1)^d \int_0^\infty \frac{\mathrm{Li}_{s-d}(-\lambda x)}{x(1+x)}\,dx.$$

Equation (5.2) proves (5.10). The same endpoint bounds give normal convergence in order, so the order derivatives are valid. At $\lambda = 1$, use $\mathcal{A}_v(1) = -\mathscr{Z}(v)$ and $\mathscr{Z}^{(r)}(0) = r(-1)^{r-1}\gamma_{r-1}$ to obtain (5.11). □

For example the choices $d = 1$ and $d = 2$ use only the kernels

$$\frac{\mathrm{Li}_{-1}(-x)}{x} = -\frac{1}{(1+x)^2}, \qquad \frac{\mathrm{Li}_{-2}(-x)}{x} = \frac{x-1}{(1+x)^3}.$$

Thus arbitrary Stieltjes indices occur in ordinary integrals with fixed rational kernels. The identity is a consequence of the proved entire-order transform; no arithmetic independence assertion is involved.

6 All mixed logarithmic and Stieltjes identities in the family

For $a \notin \{0, -1, -2, \ldots\}$ with the usual compatible branch, write

$$\zeta(1+v, a) = \frac{1}{v} + \sum_{r=0}^{\infty} \frac{(-1)^r}{r!}\gamma_r(a)v^r, \qquad \gamma_r = \gamma_r(1).$$

Only $\Re a > 0$ is needed in the formulas below.

Corollary 6.1 (Generalized Stieltjes differences)**.** *For $r \geq 0$ and $0 < \Re a < 2$,*

$$\int_0^\infty \frac{x^{-a}}{1+x}\,\partial_s^r \mathrm{Li}_s(-x)|_{s=1}\,dx = \frac{(-1)^r\pi}{\sin \pi a}\big(\gamma_r - \gamma_r(a)\big). \tag{6.1}$$

The value at $a = 1$ is removable. More explicitly it is

$$-\zeta^{(r)}(2) - r\zeta^{(r-1)}(2), \tag{6.2}$$

where the second term is omitted for $r = 0$.

Proof. Set $t = 1 - a$ in (4.10) and differentiate the entire difference $\zeta(s) - \zeta(s, a)$ r times at $s = 1$. The common pole cancels before the coefficient is taken. Lemma 4.1 justifies the differentiated integral. At $a = 1$, use (5.3) and differentiate $s\zeta(s+1)$ at $s = 1$, obtaining (6.2). □

For a particularly explicit example, the half-argument distribution formula $\zeta(s, 1/2) = (2^s - 1)\zeta(s)$ gives

$$\int_0^\infty \frac{x^{-1/2}}{1+x}\,\partial_s \mathrm{Li}_s(-x)|_{s=1}\,dx = -\pi\big(2\gamma \log 2 + \log^2 2\big). \tag{6.3}$$

Corollary 6.2 (A single Stieltjes constant at every order)**.** *For every $r \geq 1$,*

$$\boxed{\int_0^\infty \frac{x}{(1+x)^4}\,\partial_s^r \operatorname{Li}_s(-x)|_{s=3}\,dx = \frac{r(-1)^{r-1}\gamma_{r-1} - 2\zeta^{(r)}(3) - r\zeta^{(r-1)}(3)}{6}.} \tag{6.4}$$

At order zero the same integral equals $(1-2\zeta(3))/6$.

Proof. The entire normalization satisfies

$$\mathscr{Z}(v) = 1 + \sum_{r\geq 1} \frac{(-1)^{r-1}\gamma_{r-1}}{(r-1)!} v^r, \qquad \mathscr{Z}^{(r)}(0) = r(-1)^{r-1}\gamma_{r-1}.$$

Differentiate (5.9) at $s=3$. Its other entire factor has derivative $\mathscr{Z}^{(r)}(2) = 2\zeta^{(r)}(3) + r\zeta^{(r-1)}(3)$. This proves the formula, including the stated order-zero value. □

Thus the first two derivatives give the concrete ordinary integrals

$$\int_0^\infty \frac{x}{(1+x)^4}\,\partial_s \operatorname{Li}_s(-x)\big|_{s=3}\,dx = \frac{\gamma - \zeta(3) - 2\zeta'(3)}{6},$$
$$\int_0^\infty \frac{x}{(1+x)^4}\,\partial_s^2 \operatorname{Li}_s(-x)\Big|_{s=3}\,dx = -\frac{\gamma_1 + \zeta'(3) + \zeta''(3)}{3}.$$

6.1 Every power of the logarithmic integration variable

Write

$$\frac{\pi h}{\sin \pi h} = \sum_{a\geq 0} b_{2a} h^{2a}, \qquad b_0 = 1, \qquad b_{2a} = 2(1-2^{1-2a})\zeta(2a) \quad (a \geq 1).$$

For example $b_2 = \zeta(2)$ and $b_4 = 7\zeta(4)/4$.

Theorem 6.3 (All logarithmic moments)**.** *For every integer $\ell \geq 0$ and every complex s,*

$$\boxed{\int_0^\infty \frac{\log^\ell x \operatorname{Li}_s(-x)}{x(1+x)}\,dx = -\ell! \sum_{a=0}^{\lfloor \ell/2 \rfloor} b_{2a} \frac{(s)_{\ell+1-2a}}{(\ell+1-2a)!}\zeta(s+\ell+1-2a).} \tag{6.5}$$

Every product $(s)_n\zeta(s+n)$ occurring here is entire, with its removable value understood. Every fixed number of order derivatives can be taken on both sides.

Proof. The Hurwitz Taylor series at $a=1$, obtained by repeated use of $\partial_a\zeta(s,a) = -s\zeta(s+1,a)$, gives

$$\zeta(s) - \zeta(s, 1-h) = -\sum_{n\geq 1} \frac{(s)_n}{n!}\zeta(s+n)h^n, \qquad |h| < 1.$$

The putative pole at $s = 1-n$ cancels against the corresponding linear factor of $(s)_n$. The identity holds normally for s in compact sets by the holomorphy of the Hurwitz difference and Cauchy's formula. Multiply this series by $\pi/\sin\pi h$, take the coefficient of h^ℓ, and apply Lemma 4.1 to (4.10). This proves the formula and its order derivatives. □

The first two nonconstant logarithmic moments are

$$\int_0^\infty \frac{\log x \operatorname{Li}_s(-x)}{x(1+x)}\,dx = -\frac{s(s+1)}{2}\zeta(s+2),$$
$$\int_0^\infty \frac{\log^2 x \operatorname{Li}_s(-x)}{x(1+x)}\,dx = -\frac{s(s+1)(s+2)}{3}\zeta(s+3) - 2\zeta(2)s\zeta(s+1).$$

In particular,

$$\int_0^\infty \frac{\log x}{x(1+x)}\,\partial_s \operatorname{Li}_s(-x)|_{s=-1}\,dx = \frac{\gamma-1}{2}.$$

This follows by expanding $-s(s+1)\zeta(s+2)/2$ at $s=-1$; its value before differentiating is $1/2$.

For a general kernel and general scale, all Mellin derivatives at a resonant exponent have an equally finite coefficient prescription. Let

$$N_m(t,s,\lambda) = \sum_{j=0}^{m-1} c_{m,j}(t)\mathcal{H}_{m-t}(s-j,\lambda).$$

Since $N_m(k,s,\lambda)=0$, Theorem 4.3 gives

$$\partial_t^\ell \partial_s^r \mathcal{I}_m(t,s;\lambda)\Big|_{t=k} = \frac{(-1)^k \ell!}{(m-1)!}[h^{\ell+1}]\frac{\pi h}{\sin \pi h}\,\partial_s^r N_m(k+h,s,\lambda). \tag{6.6}$$

Only finitely many derivatives and coefficients are required for each fixed ℓ, r, m. This formula also retains mixed s and λ derivatives, since $\lambda\partial_\lambda$ simply shifts the order. It specifies the transverse coordinate $t-k$ and never differentiates a formula that was known only on the resonant set.

7 Exact scaled examples and further directions

Put $L = \log 2$. Classical values at $1/2$ yield

$$\mathcal{A}_1(\lambda) = \operatorname{Li}_2(1-\lambda) - \zeta(2), \tag{7.1}$$
$$\mathcal{A}_2(1/2) = -\frac{7}{4}\zeta(3) + \frac{\pi^2 L}{12} + \frac{L^3}{6}, \tag{7.2}$$
$$\mathcal{A}_2(2) = -\frac{7}{4}\zeta(3) - \frac{\pi^2 L}{4}. \tag{7.3}$$

Equation (7.1) is Euler's dilogarithm reflection identity in the present normalization. Equation (7.2) follows from the standard formulas for $\operatorname{Li}_2(1/2)$ and $\operatorname{Li}_3(1/2)$. For (7.3), the compatible boundary values are

$$\operatorname{Li}_2(2) = \frac{\pi^2}{4} - i\pi L, \qquad \operatorname{Li}_3(2) = \frac{7}{8}\zeta(3) + \frac{\pi^2 L}{4} - \frac{i\pi L^2}{2}.$$

They follow from classical inversion and the half-argument evaluations; their imaginary parts cancel in $L\operatorname{Li}_2(2) - 2\operatorname{Li}_3(2)$. The opposite compatible boundary convention gives their conjugates and the same real answer.

Theorem 5.2 consequently gives, without any integer relation search,

$$\int_0^\infty \frac{\mathrm{Li}_2(-x/2)}{x(1+x)}\,dx = -\frac{7}{4}\zeta(3) + \frac{\pi^2 L}{12} + \frac{L^3}{6}, \tag{7.4}$$

$$\int_0^\infty \frac{\mathrm{Li}_2(-2x)}{x(1+x)}\,dx = -\frac{7}{4}\zeta(3) - \frac{\pi^2 L}{4}, \tag{7.5}$$

$$\int_0^\infty \frac{\mathrm{Li}_3(-2x)}{(1+x)^3}\,dx = \frac{\pi^2(1-L)}{8} - \frac{7}{8}\zeta(3). \tag{7.6}$$

All are ordinary absolutely convergent integrals. The last example uses the short operator $(\mathsf{E}-\mathsf{E}^2)/2$ and $\mathcal{A}_1(2) = -\pi^2/4$.

The results suggest several precise continuations.

1. Determine which noninteger beta weights retain a finite expression in shifts of $\mathscr{Z}(s)$. The present factored operator uses the integrality of the denominator exponent, and does not settle finite closure for a general complex exponent.

2. Extend the factored operator to kernels with several independent negative poles. Partial fractions give a finite reduction to scaled instances of $\mathcal{A}_s$; an explicit confluent formula should keep coalescing poles and order derivatives regular simultaneously.

3. Determine which bilinear full-line integrals with two independent polylogarithm orders admit finite reductions of comparable simplicity. A product introduces an additional convolution; the present one-factor theorem does not settle that question.

4. Study the analytic continuation of the local series (5.5) around $u = \pm 2\pi i$ and its monodromy. The theorem proves normal convergence within the stated disk, but does not supply a global single-valued continuation in u.

5. Compare the precise all-order transform and resonance formulas with the older Mellin-transform tables and subsequent Euler-sum literature before asserting priority beyond the inspected corpus.

8 Completion of the undifferentiated Stieltjes edge

The positive-argument-derivative product theorem in the earlier *Polygamma Tail Identities* report [4] assumes that both ordinary derivative orders are positive. Its research question 9 asks for the missing cases in which one factor is undifferentiated. We resolve that edge. The mechanism remains the classical Mikolás product integral; the additional result is the joint spectral-pole completion, its all-index finite algorithm, and its exact endpoint law. No claim of historical priority for the classical product kernel is made. The five incoming packages at the pinned October 11 revision were also checked for duplication of this statement; their spectral Stieltjes defects concern different products.

Use the convention

$$\zeta(1+u,x) = \frac{1}{u} + \sum_{m\ge 0} \frac{(-1)^m}{m!}\gamma_m(x)u^m, \qquad \gamma_0(x) = -\psi(x).$$

Every finite part below means the constant coefficient of the integral from ϵ to 1 in the original coordinate x. Terms $\epsilon^a \log^j \epsilon$ with $a < 0$ and all positive powers of $\log \epsilon$ are deleted. A change of cutoff coordinate is not part of this convention.

8.1 The completed first-order edge

For an integer $l \geq 2$, put

$$P_l(u,v) = \frac{\Gamma(l+u+v)}{\Gamma(1+u)\Gamma(l+v)}\zeta(l+u+v), \tag{8.1}$$

$$E(u) = \frac{\pi u}{\sin \pi u}, \tag{8.2}$$

$$H_l(u,v) = -\frac{P_l(u,v)E(u) - P_l(0,v)}{u} + (-1)^l \frac{P_l(u,v)E(v) - P_l(u,0)}{v}. \tag{8.3}$$

Each divided difference in (8.3) is holomorphic near $(0,0)$. In particular H_l is a genuine two-variable Taylor germ, not an iterated choice of Laurent constants.

Theorem 8.1 (Every undifferentiated-factor jet)**.** *For all integers $m, b \geq 0$ and $l \geq 2$,*

$$\partial_u^m \partial_v^b H_l(u,v)\Big|_{u=v=0} = (-1)^m \operatorname{FP} \int_0^1 \gamma_m(x)\ \partial_t^b \zeta(t,x)\Big|_{t=l}\, dx. \tag{8.4}$$

Consequently every finite part

$$\operatorname{FP} \int_0^1 \gamma_m(x)\gamma_n^{(r)}(x)\, dx, \qquad m,n \geq 0, \quad r \geq 1,$$

has a finite expression in ordinary zeta values and derivatives at positive integers. The superscript (r) is an argument derivative.

Proof. Write $s = 1+u$ and $t = l+v$. The classical product integral continues as

$$F(s,t) = \pi \frac{\Gamma(s+t-1)}{\Gamma(s)\Gamma(t)}\zeta(s+t-1)\{\csc(\pi s) + \csc(\pi t)\}. \tag{8.5}$$

Initially it equals $\int_0^1 \zeta(s,x)\zeta(t,x)\, dx$ when $\Re s < 1$, $\Re t < 1$, and $\Re(s+t) < 1$; see Shpot–Chaudhary–Paris, Remark 1 [7], or prove it by integrating the absolutely convergent Hurwitz Fourier series first for $\Re s, \Re t < 0$. In that initial region the scalar identity $\int_0^1 \zeta(t,x)\, dx = 0$ implies

$$\int_0^1 \left(\zeta(s,x) - \frac{1}{s-1}\right)\zeta(t,x)\, dx = F(s,t).$$

Both sides have the same meromorphic continuation.

Set $R(s,x) = \zeta(s,x) - 1/(s-1)$. For s near 1, this function is holomorphic in s, and

$$R(s,x) = x^{-s} + A(s,x), \quad A(s,x) = \zeta(s,1+x) - \frac{1}{s-1}.$$

The function A is jointly holomorphic near $s=1$, $x=0$. For the second factor use

$$\zeta(t,x) = x^{-t} + B(t,x), \quad B(t,x) = \zeta(t,1+x).$$

In the product the two exponents crossing -1 at $(s,t) = (1,l)$ are precisely

$$\zeta(t)x^{-s}, \qquad c_{l-1}(s)x^{l-1-t}, \quad c_j(s) = \frac{(-1)^j(s)_j}{j!}\zeta(s+j).$$

The subtraction of $1/(s-1)$ changes only the constant Taylor coefficient of A; since $l-1 \geq 1$, the second coefficient is still $c_{l-1}(s)$ and is holomorphic near $s=1$. After subtracting sufficiently many further Taylor monomials, the remainder and all prescribed parameter derivatives are dominated by $Cx^{-\delta}(1+|\log x|^N)$, with $\delta < 1$. Their integrals therefore form a holomorphic germ.

The analytic integrals of the two resonant monomials are $-\zeta(t)/(s-1)$ and $-c_{l-1}(s)/(t-l)$, whereas all their parameter derivatives at the resonance have coordinate finite part zero: they are multiples of $x^{-1}\log^j x$. Every nonresonant monomial commutes with parameter differentiation. Thus the required holomorphic completion is

$$F(s,t) + \frac{\zeta(t)}{s-1} + \frac{c_{l-1}(s)}{t-l}.$$

Now $P_l(0,v) = \zeta(l+v)$ and $c_{l-1}(1+u) = -(-1)^l P_l(u,0)$. Substitution into (8.5) gives exactly (8.3). Finally $\partial_u^m R(1+u,x)|_{u=0} = (-1)^m \gamma_m(x)$, proving (8.4) with the stated finite-part convention. □

8.2 A terminating coefficient algorithm

Define

$$p_{r,j} = \frac{d^j}{du^j}(1+u)_r\Big|_{u=0}, \qquad p_{r,j} = 0 \text{ if } j > r.$$

For $r \geq 1$ the Hurwitz derivative law gives

$$\gamma_n^{(r)}(x) = (-1)^{n+r} \sum_{j=0}^{\min(n,r)} \binom{n}{j} p_{r,j}\, \partial_t^{n-j}\zeta(t,x)\Big|_{t=r+1}. \tag{8.6}$$

Write $P_l(u,v) = \sum_{a,b\geq 0} P_{ab}^{(l)} u^a v^b$ and $E(u) = \sum_{j\geq 0} e_{2j} u^{2j}$, where

$$e_0 = 1, \qquad e_{2j} = 2(1-2^{1-2j})\zeta(2j) \quad (j \geq 1).$$

Then

$$[u^a v^b]H_l = -\sum_{j=0}^{\lfloor (a+1)/2 \rfloor} e_{2j} P_{a+1-2j,b}^{(l)} + (-1)^l \sum_{j=0}^{\lfloor (b+1)/2 \rfloor} e_{2j} P_{a,b+1-2j}^{(l)}. \tag{8.7}$$

Combining (8.6) and (8.7) evaluates any requested m,n,r with coefficients of total degree at most $m+n+1$. There are no spectral limits left in the algorithm. The coefficient source can itself be written as

$$P_l(u,v) = \zeta(l+u+v)\prod_{j=1}^{l-1}\left(1+\frac{u}{j+v}\right)\exp\left\{\sum_{k\geq 2}\frac{(-1)^k\zeta(k)}{k}[(u+v)^k - u^k - v^k]\right\}. \tag{8.8}$$

The product is finite, and only finitely many terms of the exponential can affect a prescribed coefficient. This exhibits the claimed ordinary-zeta closure directly. In conjunction with the inherited positive-positive sector this proves a useful uniform statement: whenever the total ordinary derivative order is positive, every quadratic same-point Stieltjes finite part reduces to a finite expression involving ordinary positive-integer zeta values and their order derivatives. No Stieltjes constant is needed in those final expressions. The double-undifferentiated corner is different and is recorded below for comparison.

For completeness the edge and inherited interior can be written in one finite dictionary. For $r, t \geq 0$ with $r + t > 0$, put $k = r + 1$, $l = t + 1$, $w = r + t + 1$, and define

$$P_{kl}(u, v) = \frac{\Gamma(w + u + v)}{\Gamma(k + u)\Gamma(l + v)}\zeta(w + u + v),$$
$$H_{kl}(u, v) = (-1)^k \frac{P_{kl}(u, v)E(u) - P_{kl}(0, v)}{u} + (-1)^l \frac{P_{kl}(u, v)E(v) - P_{kl}(u, 0)}{v}. \tag{8.9}$$

Then the uniform coefficient formula is

$$\boxed{\mathrm{FP} \int_0^1 \gamma_m^{(r)}(x)\gamma_n^{(t)}(x)\, dx = (-1)^{m+n+r+t} m!n![u^m v^n](1 + u)_r(1 + v)_t H_{kl}(u, v).} \tag{8.10}$$

When $r = 0$ this is the new edge theorem; when $t = 0$ use symmetry. When both are positive it is the inherited positive-positive formula [4, Theorem `hprod:stieltjes-products`]. Multiplying its Hurwitz Taylor germ by the two rising factorials proves (8.10). Only order derivatives at the single argument $w = r + t + 1$ occur. Their coefficients involve rational numbers and ordinary zeta values from the Gamma expansion. Thus "single argument" refers to the differentiated zeta factors, not to every undifferentiated zeta value in the coefficients.

8.3 Explicit first-index identities

Let $H = H_{l-1}$, $H_2 = H_{l-1}^{(2)}$, and $\sigma = (-1)^l$. The first coefficients are

$$H_l(0, 0) = -H\zeta(l) + (\sigma - 1)\zeta'(l), \tag{8.11}$$
$$\partial_u H_l(0, 0) = \left[(\sigma - 1)\zeta(2) + \left(\frac{1}{2} - \sigma\right) H_2 - \frac{1}{2}H^2\right] \zeta(l) + (\sigma - 1)H\zeta'(l) + \left(\sigma - \frac{1}{2}\right) \zeta''(l), \tag{8.12}$$
$$\partial_v H_l(0, 0) = \{(\sigma - 1)\zeta(2) + H_2\}\zeta(l) - H\zeta'(l) + \left(\frac{\sigma}{2} - 1\right) \zeta''(l). \tag{8.13}$$

For example the $l = 2$ formulas give the pair

$$\boxed{\mathrm{FP} \int_0^1 \gamma_1(x)\gamma_0'(x)\, dx = \frac{1}{2}\zeta''(2) - \zeta(2),} \tag{8.14}$$

$$\boxed{\mathrm{FP} \int_0^1 \gamma_0(x)\gamma_1'(x)\, dx = -\zeta'(2) - \frac{1}{2}\zeta''(2).} \tag{8.15}$$

Neither individual identity follows merely by integrating the derivative of $\gamma_0\gamma_1$. Their sum is the endpoint constant $-\zeta(2) - \zeta'(2)$. At one further argument derivative,

$$\mathrm{FP} \int_0^1 \gamma_1(x)\gamma_0''(x)\, dx = \left(4\zeta(2) - \frac{3}{2}\right) \zeta(3) + 6\zeta'(3) + 3\zeta''(3). \tag{8.16}$$

One higher Stieltjes index already produces a new derivative order:

$$\mathrm{FP} \int_0^1 \gamma_2(x)\gamma_0'(x)\, dx = 2\zeta'(2) - \zeta''(2) - \frac{2}{3}\zeta'''(2) + 2\zeta(2)\zeta(3). \tag{8.17}$$

More generally, put $J_{0,1}^{m,n} = \mathrm{FP} \int_0^1 \gamma_m(x)\gamma_n'(x)\, dx$. In its finite formula the coefficient of the highest ordinary-zeta order derivative $\zeta^{(m+n+1)}(2)$ is exactly

$$\frac{(-1)^{m+n+1}(m - n)}{(m + 1)(n + 1)}. \tag{8.18}$$

This is a coefficient statement for the displayed reduction and does not assert arithmetic independence of its evaluated constants. Indeed the highest derivative can only come from $-P^{(2)}_{m+1,n} + P^{(2)}_{m,n+1}$ with all Gamma and Pochhammer factors replaced by their constant terms. Multiplication by $(-1)^{m+n+1}m!n!$ gives (8.18).

For readers preferring an ordinary convergent integral, put $g(x) = \gamma_1(1+x)$ and $h(x) = \psi'(1+x)$. Equation (8.14) is equivalent to

$$\int_0^1 \left\{ \frac{g(x) - \gamma_1 - (\zeta(2) + \zeta'(2))x}{x^2} + \frac{\log x}{x}\{h(x) - \zeta(2)\} + g(x)h(x) \right\} dx = \zeta(2) - \frac{1}{2}\zeta''(2) + \frac{1}{4} + \gamma_1. \tag{8.19}$$

The first apparent singularity is removable and the second is $O(\log x)$. Indeed $\gamma_1(x) = \log x/x + g(x)$ and $\psi'(x) = x^{-2} + h(x)$; the removed singular monomials have finite-part integral $-1/4 - \gamma_1$, since those proportional to x^{-1} or $x^{-1}\log x$ have finite part zero. This proves (8.19) without any numerical limiting prescription.

8.4 All endpoint constants and compatibility with integration by parts

For $r, t, m, n \geq 0$ write

$$J^{m,n}_{r,t} = \mathrm{FP}\int_0^1 \gamma^{(r)}_m(x)\gamma^{(t)}_n(x)\,dx.$$

Set $p_{0,0} = 1$ and $p_{0,j} = 0$ for $j > 0$, consistently with $(1+u)_0 = 1$.

Theorem 8.2 (Exact endpoint law)**.** *For every $r, t, m, n \geq 0$,*

$$\boxed{\begin{aligned} J^{m,n}_{r+1,t} + J^{m,n}_{r,t+1} = &-\frac{(-1)^{m+r}p_{r,m}}{(r+1)!}\gamma^{(r+t+1)}_n(1) \\ &-\frac{(-1)^{n+t}p_{t,n}}{(t+1)!}\gamma^{(r+t+1)}_m(1). \end{aligned}} \tag{8.20}$$

All constants on the right reduce to positive-integer ordinary-zeta jets by (8.6).

Proof. The shift identity gives $\gamma_m(x) = x^{-1}\log^m x + \gamma_m(1+x)$ exactly. Its rth derivative has singular part

$$S_{m,r}(x) = \frac{(-1)^r}{x^{r+1}}\sum_{j=0}^{\min(m,r)}(-1)^j\binom{m}{j}p_{r,j}\log^{m-j} x.$$

The coefficient of $x^{-r-1}\log^0 x$ is $(-1)^{m+r}p_{r,m}$; it vanishes if $m > r$. The smooth factor is $\gamma^{(r)}_m(1+x)$. The constant term of the product at zero has three contributions: the product of its two smooth constant terms, and the two products of a singular logarithm-free coefficient with the matching smooth Taylor coefficient. The singular–singular product has power x^{-r-t-2} and contributes no constant. The first contribution cancels the endpoint value at $x = 1$. The remaining two are exactly the negative of the right side of (8.20). Apply the fundamental theorem of calculus on $[\epsilon, 1]$ and take constant coefficients. □

This law provides a useful consistency check on the completed kernels. It also shows precisely why the first-derivative edge contains information beyond the old positive-positive sector: integration by parts determines $J^{m,n}_{0,1} + J^{m,n}_{1,0}$, but not its antisymmetric part. Once the first edge has been

established, higher edges can also be related to the positive-positive formulas while retaining the explicit endpoint terms. In particular no formal "integral of a derivative is zero" rule is appropriate here.

8.5 The double-undifferentiated corner

The corner $r = t = 0$ is already treated in the manuscript's coincident-endpoint Stieltjes chapter. For compatibility, a useful holomorphic presentation is as follows. Define

$$A(u) = u\zeta(1+u), \tag{8.21}$$

$$C(u,v) = \frac{\Gamma(1+u+v)}{\Gamma(1+u)\Gamma(1+v)} \frac{\pi uv\{\csc(\pi u) + \csc(\pi v)\}}{u+v} A(u+v). \tag{8.22}$$

All apparent singularities in C are removable near zero. In particular $C(u,0) = A(u)$ and $C(0,v) = A(v)$. The completed corner is

$$H_{11}(u,v) = \frac{A(u) + A(v) - 1 - C(u,v)}{uv}. \tag{8.23}$$

Its Taylor coefficients satisfy $J_{0,0}^{m,n} = (-1)^{m+n} m!n![u^m v^n]H_{11}$. This expression follows from the same local subtraction, now applied to both factors $\zeta(1+u,x) - 1/u$ and $\zeta(1+v,x) - 1/v$: the integrated spectral product is $F(1+u,1+v) + 1/(uv)$, and the local resonance corrections are $(\zeta(1+v) - 1/v)/u$ and $(\zeta(1+u) - 1/u)/v$. The two pole subtractions must be made jointly. Equation (8.23) records compatibility with the inherited corner rather than claiming it as another new result.

8.6 Independent verification

The supplied script constructs the local Gamma germ with exact rational arithmetic and formal ordinary-zeta derivatives. It checks all 81 endpoint identities with $0 \le m, n, r, t \le 2$, compares the printed low-index identities, and exports their formulas. This is an exact finite verification of the implementation, not a replacement for the analytic proof at all indices. The numerical suite computes the left sides by integrating their convergent local Laurent–logarithmic expansions up to a positive splitting point and then using ordinary quadrature to 1. Thus it does not evaluate the product kernel to obtain both sides of a test. Its precision, truncation order, actual residuals, and second splitting-point check are recorded in the JSON output. These arbitrary-precision comparisons are diagnostics rather than interval certificates.

9 Complete optimal reconstruction at smaller tail order ten

Two recent source theorems leave only one pair at smaller order ten. The order-ten theorem of *Alternating Tails and Spectral Defects*, Theorem `rt:rank10`, proves the conjecture for every integer $b \ge 6$ outside the progression $b \equiv 962 \pmod{2048}$. The effective theorem of *Quartic Spectral Reconstruction*, Theorem `ert:main` and its exact threshold calculation, proves it for every $b \ge 1920$. We certify the single remaining value $b = 962$ by a five-square determinant over a finite field. This completes the whole smaller-order-ten family. The method does not assert that the unrestricted unequal-order conjecture is proved.

9.1 The formal coordinate problem

For even p, q put

$$A_p(n) = \zeta(p, n+1) = \zeta(p) - H_n^{(p)}, \qquad x_n = n + \tfrac{1}{2},$$

and initially define

$$D_{p,q}(s) = \sum_{n\geq 0} A_p(n) A_q(n) x_n^{-s}, \qquad \Re s > 3 - p - q.$$

Let $D_r = D_{p,q}(-2r)$ denote its regular continued centered moments. The centered Hurwitz expansion has powers x_n^{1-p-2j}; hence the product has only even inverse powers. Its possible continuation poles lie at odd integers, so all the displayed $-2r$ values are regular. The inherited centered continuation and finite Bernoulli reduction supply inverse transforms

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_r, \tag{9.1}$$

$$D_r = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) C_j. \tag{9.2}$$

Write $p = 2a$, $q = 2b$, $a < b$. To specify the algebraic claim completely, let $\mathcal{V}_{a,b}$ be the $2b$-dimensional rational vector space with *formal basis*

$$C_0, \ldots, C_{b-1}, Z_0, \ldots, Z_{b-1}.$$

These are independent labels by definition. Under numerical evaluation, Z_k is assigned $\zeta(2k)$, including $Z_0 = -1/2$, and each early C_j is assigned its continued value above. No injectivity of this evaluation map is asserted. For every $j \geq b$, the source reduction defines $C_j = \sum_{k=0}^{b-1} M_{j-b,k} Z_k$, using the formula below also for row indices greater than $b-1$. Thus every later D_r is a specified vector of $\mathcal{V}_{a,b}$. The coordinate matrix of $C_0, \ldots, C_{2b-1}$ is $\operatorname{diag}(I_b, M_{a,b})$. It follows that the first $2b$ moment coordinates have full formal rank precisely when the matrix

$$(M_{a,b})_{ik} = B_{2(i+k-a+1)} \left\{ \frac{\binom{2i+2}{2a-2k}}{2i+2} + \frac{\binom{2b+2i-2a+2}{2b-2k}}{2b+2i-2a+2} \right\}, \qquad 0 \leq i, k < b, \tag{9.3}$$

is nonsingular. A negative Bernoulli index gives zero. A binomial summand is omitted unless its lower index is at least two and no greater than its upper index. Thus a lower index zero is omitted. The complete first-$2b$ coordinate determinant is $\det M_{a,b}/(4b-1)!!$. These are inherited identities, not new independence assertions about numerical zeta values.

Theorem 9.1 (Complete smaller-order-ten reconstruction)**.** *For every integer $b \geq 6$, $M_{5,b}$ is nonsingular. Consequently, for every even $q > 10$, the moments*

$$D_{10,q}(0), D_{10,q}(-2), \ldots, D_{10,q}(-2q+2)$$

have formal coordinate rank q in the inherited coordinate space. Every later centered moment is a finite rational linear combination of these first q moments.

The only new nonvanishing certificate needed by the theorem is

$$\boxed{\det M_{5,962} \equiv 15578 \pmod{65537}.} \tag{9.4}$$

The rational determinant has denominator prime to 65537; therefore this congruence proves exact nonvanishing.

9.2 The retained five-square matrix

We state the inherited contraction explicitly to make the finite certificate independently reproducible. For $b \geq 2a$ and $0 \leq i, \ell < a$, define

$$n_i = b - a + i, \quad N_{i\ell} = 2(n_i - \ell) + 1, \quad R_\ell = 2b - 2a + 2 + 2\ell,$$

$$F_{i\ell} = \frac{1}{(2\ell+1)!} \prod_{r=0}^{2\ell-1} (2n_i - r)(2n_i + 3 + r).$$

Define the integer Newton coefficients d_{ih} by

$$(2n_i + 2 - y)(4n_i + 2 - y)^{\underline{2i}} = \sum_{h=0}^{2i+1} d_{ih} y^{\underline{h}}. \tag{9.5}$$

Here $z^{\underline{h}} = z(z-1)\cdots(z-h+1)$. Equivalently, if the polynomial on the left is $W_i(y)$, then

$$d_{ih} = \frac{1}{h!} \sum_{j=0}^{h} (-1)^{h-j} \binom{h}{j} W_i(j).$$

The retained matrix is

$$\mathcal{A}_{i\ell} = 2F_{i\ell} + \frac{R_\ell!}{(2n_i+1)(2n_i+2)(2\ell+2)! N_{i\ell}!} \sum_{k=0}^{i} d_{i,2k+1} N_{i\ell}^{\underline{2k+1}} 2B_{2b-2(a-i+\ell+k)}. \tag{9.6}$$

The omitted lower-index-zero convention in (9.3) is already included in the $2F$ term and must not be altered.

For completeness, the source multiplier is

$$\begin{aligned} \det M_{a,b} = (-1)^{a(b-a)+a(a-1)/2} & \frac{\prod_{t=0}^{b-1}(2b+2t-2a+1)!}{\prod_{k=0}^{b-1}(2b-2k)!} \\ & \cdot \prod_{n=0}^{b-a-1} \frac{((2n+1)!)^2}{(4n+2)!(4n+3)!} \prod_{i=0}^{a-1} \frac{(2n_i+1)(2n_i+2)!}{(4n_i+2)!} \det \mathcal{A}. \end{aligned} \tag{9.7}$$

The exact derivation is the Padé-row elimination in the source's Lemma `ert:normalization`. It can also be replayed directly: the monic rows

$$p_{nt} = \frac{2(2n+2t+1)!(2n+1)!}{(4n+2)!(2n-2t+1)!(2t)!}, \qquad 0 \leq t \leq n,$$

are applied to the factorial-normalized Bernoulli matrix. Their orthogonality eliminates the last $b-a$ columns from the final a rows. Reversing the retained columns, and dividing row i by $c_{n_i} = (2n_i+1)(2n_i+2)!/(4n_i+2)!$, gives (9.6). The diagonal eliminated entries are $((2n+1)!)^2/((4n+2)!(4n+3)!)$, which gives (9.7), including its sign.

9.3 The complete finite certificate

Set $a = 5$, $b = 962$, and $P = 65537$. The prime P exceeds every factorial argument in (9.6) and (9.7). The Bernoulli recurrence

$$B_0 = 1, \qquad B_n = -\frac{1}{n+1} \sum_{j=0}^{n-1} \binom{n+1}{j} B_j \quad (n \geq 1) \tag{9.8}$$

shows that every B_n used here has denominator prime to P: all indices are at most $1922 < P-1$. Thus the entire calculation can be performed in $\mathbb{F}_P$ without an undefined division. Using $B_1 = -1/2$ and $B_{2r+1} = 0$ for $r \geq 1$, the nine required residues, in descending-index order, are

$2r$	1922	1920	1918	1916	1914	1912	1910	1908	1906
$B_{2r} \bmod P$	31528	45962	26974	45003	39580	2844	23974	18828	26318.

(9.9)

Substitution into (9.6) gives

$$\mathcal{A} \equiv \begin{pmatrix} 25959 & 60986 & 10404 & 1273 & 2935 \\ 16142 & 62424 & 8365 & 2781 & 39264 \\ 13624 & 13203 & 159 & 63417 & 38163 \\ 51637 & 50007 & 47664 & 40198 & 52007 \\ 23028 & 22542 & 12445 & 19571 & 47569 \end{pmatrix} \pmod{65537}. \tag{9.10}$$

Gaussian elimination without row exchanges has diagonal pivots

$$25959, \quad 2542, \quad 6997, \quad 61613, \quad 26239.$$

Their product is 60407 modulo 65537. The complete multiplier in (9.7), including its sign, is 49105 modulo 65537. Therefore

$$\det M_{5,962} \equiv 49105 \cdot 60407 \equiv 15578 \not\equiv 0 \pmod{65537},$$

which proves (9.4).

Proof of Theorem 9.1. If $b \not\equiv 962 \pmod{2048}$, use the source order-ten theorem. If $b \geq 1920$, use the source effective eventual theorem with $B_5 = 1920$. The only integer b with $6 \leq b < 1920$ and $b \equiv 962 \pmod{2048}$ is 962; (9.4) treats that value. These cases cover all integers $b \geq 6$. The inherited triangular transforms and finite late-coordinate expansion then give the stated moment reconstruction. □

The supplied verifier `complete_p10.py` implements (9.8), (9.5), (9.6), and modular Gaussian elimination, using only Python integer arithmetic. A second verifier reconstructs the same matrix from the uncontracted Padé row sums and separately reduces exact rational Bernoulli numbers obtained from SymPy. These checks agree entry by entry. The prime is checked by trial division through its integer square root. No floating-point determinant is used anywhere in the proof. This is a finite arithmetic proof certificate, not a claim of proof-assistant formalization.

9.4 An ordinary convergent polygamma identity for every larger order

The nonvanishing theorem makes the inherited reconstruction an unconditional all-q identity family at smaller order ten. Define, for $0 \leq k < b$,

$$P_{b,k}(x) = \sum_{t=0}^{b-1} (M_{5,b}^{-1})_{kt} \left\{ (x+\tfrac{1}{2})^{2b+2t+1} - (x-\tfrac{1}{2})^{2b+2t+1} \right\} = \sum_{r=0}^{2b-1} w_{b,k,r} x^{2r}. \tag{9.11}$$

All coefficients are rational. Put

$$s_{p,0} = \frac{1}{p-1}, \qquad s_{p,j} = \frac{B_{2j}(1/2)}{(2j)!}(p)_{2j-1} \quad (j \geq 1), \qquad \alpha_j = \sum_{h=0}^{j} s_{10,h} s_{2b,j-h},$$

where $(p)_m$ is the rising factorial. Define the finite even polynomial

$$S_{b,k}(x) = \sum_{r=b+4}^{2b-1} w_{b,k,r} \sum_{j=0}^{r-b-4} \alpha_j x^{2(r-b-4-j)}. \tag{9.12}$$

Then

$$\boxed{\sum_{n=0}^{\infty} \left\{ P_{b,k}(x_n) \, \frac{\psi^{(9)}(n+1)\psi^{(2b-1)}(n+1)}{9!\,(2b-1)!} - S_{b,k}(x_n) \right\} = \zeta(2k).} \tag{9.13}$$

Every sum in (9.13) converges absolutely. Indeed, the centered Hurwitz expansion is

$$A_p(n) \sim \sum_{j \geq 0} s_{p,j} x_n^{1-p-2j}.$$

Thus (9.12) removes exactly every nondecaying term of $P_{b,k}(x_n)A_{10}(n)A_{2b}(n)$ and leaves $O(x_n^{-2})$. The polynomial difference in (9.11) is exactly the inverse transform (9.1). Matrix inversion therefore makes its continued sum $\zeta(2k)$. All removed monomials have zero centered continued sum, since $\zeta(-2j, 1/2) = 0$ for integers $j \geq 0$, proving the ordinary convergent identity. The polygamma substitution uses $\psi^{(p-1)}(n+1) = (p-1)!\zeta(p, n+1)$ for even p.

The new contribution is the complete smaller-order-ten theorem, including its single missing exact certificate. The centered continuation, Padé contraction, and conversion of reconstructed coordinates into convergent sums retain their source attribution.

10 Complete optimal reconstruction at smaller tail order twelve

Research question R1 of the source *Quartic Spectral Reconstruction* explicitly asks for complete smaller-order-ten and smaller-order-twelve families. The preceding section and the theorem below resolve those two specified tasks. At smaller order twelve, the preceding one-case synthesis is unavailable: no source theorem handles almost every parameter. We instead prove an explicit eventual threshold by an elementary coefficient estimate, then certify the entire remaining finite set over one finite field. The proof uses only six-square retained matrices, even when the original matrix has more than eleven thousand rows.

Theorem 10.1 (Complete smaller-order-twelve reconstruction)**.** *For every integer $b \geq 7$, the matrix $M_{6,b}$ defined in (9.3) is nonsingular. Consequently, for every even $q > 12$, the first q centered moments of $A_{12}(n)A_q(n)$ have formal coordinate rank q, and every later centered moment is a finite rational linear combination of them.*

The proof has two exact components:

$$\det M_{6,b} \neq 0 \quad (b \geq 11520), \qquad \det M_{6,b} \neq 0 \quad (7 \leq b \leq 11519).$$

The first component has an ordinary analytic proof below. The second is a finite arithmetic certificate with 11513 cases, reproduced in full by two independently implemented verifiers. No sampling of the parameter range is used.

10.1 A six-square polynomial and its leading determinant

In (9.5)–(9.6), set $a = 6$ and replace $2B_{2b-2j}$ by independent indeterminates U_j, $1 \le j \le 11$. Write the resulting six-square matrix as $\mathcal{A}(b; U)$. Set

$$\begin{aligned} n_i &= b - 6 + i, & R_\ell &= 2b - 10 + 2\ell, \\ J(b) &= \prod_{i=0}^{5} (2n_i + 1)(2n_i + 2), & E(b) &= \prod_{i,\ell=0}^{5} \prod_{r=1}^{\max(0,2i-4\ell-1)} (R_\ell + r), \\ D(b) &= 12! J(b) E(b), & \mathscr{P}(b; U) &= \det(D(b)\mathcal{A}(b; U)). \end{aligned}$$

The polynomial E has 35 linear factors. The source factorial cancellation proves

$$D\mathcal{A} \in \mathbb{Z}[b, U_1, \dots, U_{11}]^{6\times 6}, \qquad \deg_b(D\mathcal{A}_{i\ell}) \le 67, \qquad \deg_b \mathscr{P} \le 402.$$

Indeed $N_{i\ell} - R_\ell = 2i - 4\ell - 1$, so E clears exactly the possible denominator products in $R_\ell!/N_{i\ell}!$; J clears the row factors and 12! clears the fixed factorials. All these factors are positive when $b \ge 12$.

Let $\mathscr{P}_{[6]}$ denote the part homogeneous of degree six in the U variables. The inherited highest-degree determinant identity, specialized to six, gives at $U_j = 2B_{2b-2j}$

$$\left|\mathscr{P}_{[6]}\right| = \frac{\mathcal{G}(b)(4F_0)^6}{(2\pi)^{12(b-6)}} \det\big(\zeta(s_0 + 2i + 2j)\big)_{0\le i,j<6}, \qquad s_0 = 2b - 22, \quad F_0 = s_0!, \tag{10.1}$$

where

$$\begin{aligned} \mathcal{G}(b) &= \kappa J(b)^5 E(b)^6 \prod_{i=0}^{5} (4n_i + 2)^{\underline{2i}} \prod_{\ell=0}^{5} \prod_{r=1}^{12+2\ell} (2b - 22 + r), \\ \kappa &= \frac{6!(12!)^6}{\prod_{\ell=0}^{5}(2\ell + 2)!} = 359054212827003991621632000000 0. \end{aligned}$$

The degree of $\mathcal{G}$ is 402, and every linear factor displayed in it is at least b when $b \ge 24$. Hence

$$\mathcal{G}(b) \ge \kappa b^{402} \qquad (b \ge 24). \tag{10.2}$$

Here the leading determinant is a structural identity, not an estimate of an expanded polynomial. Its short derivation uses $N_{i\ell} - (2k+1) = 2b - 2(6 - i + \ell + k)$ in (9.6) to cancel factorials. The resulting linear-in-U part factors through the matrix $(V_{6-i+\ell})$, where $V_j = U_j F_0/(2b - 2j)!$. The remaining row transformation is triangular, with diagonal $d_{i1} = -(i + 1)(4n_i + 2)^{\underline{2i}}$. Taking determinants, reversing rows, and applying Euler's formula $2B_{2m} = (-1)^{m-1} 4(2m)!\zeta(2m)/(2\pi)^{2m}$ gives (10.1) with the displayed $\mathcal{G}$. This retains the attribution of the general highest-degree factorization in the source *Quartic Spectral Reconstruction*.

The zeta determinant in (10.1) is a positive moment Gram determinant: use the points r^{-2} with weights r^{-s_0}, $r \ge 1$. Cauchy–Binet and the first six distinct support points give

$$\begin{aligned} \det\big(\zeta(s_0 + 2i + 2j)\big)_{0\le i,j<6} &\ge V\, 720^{-s_0}, \\ V &= \prod_{1\le r<s\le 6} (r^{-2} - s^{-2})^2 \\ &= \frac{14235529}{37439062426244874240000000000000} > 0. \end{aligned} \tag{10.3}$$

In particular, the highest-degree component cannot vanish.

10.2 A simple coefficient estimate requiring no expanded determinant

For an integer polynomial let $\|\cdot\|_1$ be the sum of absolute values of its coefficients. It is submultiplicative. We shall use the deliberately generous bound

$$\max_{0\le i,\ell<6}\|D\mathcal{A}_{i\ell}\|_1<2^{512}. \tag{10.4}$$

The following elementary factor count proves it. Every factor in J has norm less than 16, and every factor in E has norm less than 16. Thus

$$\|J\|_1<2^{48},\qquad \|E\|_1<2^{140},\qquad 12!<2^{29}.$$

After removing the two factors for one fixed row, the remaining J product has norm less than 2^{40}. After cancellation in $ER_\ell!/N_{i\ell}!$, at most 21 additional linear factors are present, each of norm less than 32. Its norm is therefore less than 2^{245}. Also

$$\|N_{i\ell}^{h}\|_1<2^{55}\quad(1\le h\le 11),\qquad \|d_{ih}\|_1<2^{65}.$$

For the last assertion, apply the Newton formula to

$$W_i(j)=(2b-10+2i-j)\prod_{r=0}^{2i-1}(4b-22+4i-j-r),\qquad 0\le j\le h\le 2i+1\le 11.$$

Its first factor has norm below 16 and each remaining factor has norm below 32, so $\|W_i(j)\|_1<2^{54}$. The sum of the binomial coefficients in the Newton formula is $2^h\le 2^{11}$; division by $h!$ only decreases the bound.

The numerator defining $F_{i\ell}$ has at most 20 linear factors, each of norm below 32. Thus the constant term $2DF_{i\ell}$ has norm below 2^{318}. Each coefficient of an indeterminate U_j in $D\mathcal{A}_{i\ell}$ has norm below

$$2^{29+40+245+65+55}=2^{434}.$$

There are at most six such coefficients. Consequently the entry norm is below $2^{318}+6\cdot 2^{434}<2^{437}<2^{512}$, proving (10.4). This argument also proves integrality by the same explicit cancellations; no fitted coefficient bound is involved. A supplementary exact script also computes a sharper factor-product bound; the proof uses only the elementary bound just established.

The determinant coefficient norm is at most

$$K=6!(2^{512})^6=720\,2^{3072}.$$

For $b\ge 11520$ put

$$T=\frac{8(2b)^{20}F_0}{(2\pi)^{s_0}}>1.$$

The inequality $T>1$ follows from $s_0!/(2\pi)^{s_0}>(s_0/24)^{s_0}>1$, using $s_0!>(s_0/3)^{s_0}$ and $2\pi<8$. Euler's Bernoulli formula and $\zeta(2m)<2$ show $|2B_{2b-2j}|\le T$ for $1\le j\le 11$. Since all remaining terms of $\mathscr{P}$ have total U degree at most five,

$$|\mathscr{P}_{[\le 5]}|\le Kb^{402}T^5.$$

Combining this estimate with (10.1)–(10.3), then using $2\pi<8$, gives the explicit domination ratio

$$\frac{|\mathscr{P}_{[\le 5]}|}{|\mathscr{P}_{[6]}|}\le C\,\frac{b^{100}5760^{2b}}{(2b-22)!},\qquad C=\frac{2^{3289}}{\kappa V\,720^{21}}<2^{3289}. \tag{10.5}$$

The final inequality follows, for example, from the exact rational value

$$\kappa V=\frac{786377640075797}{57600000}>1.$$

10.3 A threshold of 11520 proved by elementary inequalities

Define

$$R(b) = 2^{3289}\frac{b^{100}5760^{2b}}{(2b-22)!}.$$

We prove $R(b) < 1$ for every integer $b \geq 11520$. At $b = 11520$, write $n = 23018$. The elementary factorial estimate $n! \geq (n/e)^n > (n/3)^n$, obtained by integrating $\log x$, gives

$$n! > 7200^n = 5760^n(5/4)^n,$$

since $n > 3 \cdot 7200$. Moreover $11520 < 2^{14}$, $5760 < 2^{13}$, and $(5/4)^4 = 625/256 > 2$. Therefore

$$R(11520) < 2^{3289+1400+286}(4/5)^{23018} < 2^{4975-5754} = 2^{-779} < 1. \tag{10.6}$$

For every subsequent integer b,

$$\frac{R(b+1)}{R(b)} = \left(1+\frac{1}{b}\right)^{100}\frac{5760^2}{(2b-20)(2b-21)} < \frac{3}{4},$$

because $b \geq 100$ gives the first factor less than 3, both denominator factors exceed b, and $b \geq 11520 = 2 \cdot 5760$. Thus the ratio in (10.5) is strictly below one for all $b \geq 11520$. Its highest-degree component cannot be cancelled. As $D(b) > 0$ and the source multiplier (9.7) is nonzero, this proves $\det M_{6,b} \neq 0$ throughout that infinite range. The explicit threshold follows from a mathematical argument, not from the observed behavior of a finite list of determinants.

10.4 The complete finite-field certificate below the threshold

For the remaining 11513 integers $7 \leq b \leq 11519$, use the single prime $P = 65537$. The original source matrix is used for $7 \leq b \leq 11$, where the contraction is not asserted. For $12 \leq b \leq 11519$, use $\mathcal{A}$ from (9.6) with $a = 6$. The source relation (9.7) implies that nonvanishing of this six-square determinant suffices.

Every division is defined modulo P. The largest Bernoulli index needed for the retained matrices is $2 \cdot 11519 - 2 = 23036 < P - 1$; (9.8) therefore puts all these Bernoulli numbers in $\mathbb{Z}_{(P)}$. Every factorial argument and every positive linear denominator appearing in the matrices and the exact source multipliers is less than $4 \cdot 11520 + 5 = 46085 < P$. In particular, the entire source multiplier is a unit modulo P. The computation never replaces a singular reduction modulo P by an unsupported claim of rational nonvanishing: every recorded residue is nonzero.

The file `complete_p12.json` records the full ordered residue list, with its first entry corresponding to $b = 7$. It has 11513 entries, and every entry belongs to $\{1, \ldots, 65536\}$. For orientation, a few entries are

b	7	8	11	12	100	962	4096	11519
residue	39450	1561	25713	60062	43034	42605	25983	23212.

(10.7)

For $b \leq 11$ the entry is $\det M_{6,b} \bmod P$; for $b \geq 12$ it is $\det \mathcal{A} \bmod P$. The displayed sample is explanatory; the proof checks the complete ordered list.

Two full implementations replay the certificate by different exact algorithms. The C++ verifier uses the triangular Bernoulli recurrence, finite differences for the Newton coefficients at each integer b, and modular Gaussian elimination. The Python verifier instead inverts $(e^t - 1)/t$ as a truncated

power series, using exact integer Kronecker multiplication. This produces $B_n/n!$ directly. It then uses the factorial-cancelled entry formula

$$\mathcal{A}_{i\ell} = 2F_{i\ell} + \frac{R_\ell!}{(2n_i+1)(2n_i+2)(2\ell+2)!}\sum_{k=0}^{i} d_{i,2k+1}\frac{2B_{2b-2(6-i+\ell+k)}}{(2b-2(6-i+\ell+k))!}, \tag{10.8}$$

with Newton coefficients computed as global integer polynomials in b, and evaluates determinants by a division-free subset Laplace expansion. It verifies every one of the 11513 residues, checks that they are all nonzero, and compares the complete result with the independently produced C++ list. The C++ calculation also compares the original and contracted determinant formulas throughout $12 \le b \le 32$, including the full exact multiplier.

The exact polynomial multiplication in the Python verifier represents coefficients as successive base-2^{64} digits of an integer. At the largest product length here, every unreduced convolution coefficient is smaller than 2^{64}; hence there is no carry between coefficient blocks. Integer multiplication is therefore precisely polynomial convolution, with no floating-point transform or rounding. Newton inversion is checked by multiplying the final series by $(e^t-1)/t$ through degree 23039 and obtaining the constant polynomial one. Both complete replays use finite integer arithmetic. Together with (10.6), they prove Theorem 10.1.

10.5 Identity consequences and the remaining problem

The new rank theorem yields the ordinary convergent identity family (9.13) with the following replacements: $M_{5,b}$ by $M_{6,b}$, 10 by 12, 9! by 11!, and $\psi^{(9)}$ by $\psi^{(11)}$. In the subtraction polynomial, replace the lower outer index $b+4$ by $b+5$ and each occurrence of $r-b-4$ by $r-b-5$. The allowed range is $b \ge 7$. Thus products of the eleventh polygamma with every higher odd-order polygamma now reconstruct all b even-zeta coordinates by explicit rational polynomial weights and finite midpoint subtractions.

Theorems 9.1 and 10.1 extend the proved smaller orders from $2, 4, 6, 8$ to $10, 12$. The full fixed-gap-six family and the unrestricted unequal-order conjecture remain open. The eventual theorem reduces each further fixed smaller order to a finite certificate, but no assertion about uncomputed orders is made here. The leading Hankel factorization and the source contraction are inherited; the completed order-twelve family, its simple explicit threshold proof, and its full pair of arithmetic certificates are the new contributions of this section.

11 Source reconciliation, verification, and further research

11.1 Precise changes to the research status

The targeted audit has not established an additional false theorem in the inspected source dependencies. The proposed corrections concern the scope and status of research questions, the distinction between inherited and new results, and the conventions needed when combining formulas. This distinction matters in a rapidly developing collection: the date on an archive does not by itself determine which other results its research list has incorporated.

The reconstruction questions. Question R1 in the latest *Quartic Spectral Reconstruction* report asks for the complete smaller-order-ten family and then the smaller-order-twelve family [3]. Those two specified tasks are now resolved by Theorems 9.1 and 10.1. In particular, the older excluded

progression for order ten is no longer an unresolved family: the inherited eventual theorem and congruence theorem reduce the remaining task to the single case $b = 962$, which is certified here. For order twelve, the complete finite range and the proved eventual range cover every $b \geq 7$. The unrestricted assertion for every $a < b$, requested in R2 of the quartic report, remains open. So does the full fixed-gap-six family $b = a + 3$. The newly included cases $(p, q) = (10, 16)$ and $(12, 18)$ do not settle that family. Every rank statement is in the specified formal coordinate space; no arithmetic independence of zeta values follows from it.

The alternating-tail question. The question headed "Bilinear adjacent-tail certificates" in the latest alternating-tail report [2] is answered within exactly its two-product rational ansatz. Theorem 2.1 gives necessity and sufficiency, and Theorem 2.3 supplies the finite and infinite endpoint identities. The independently shifted quadratic test case is therefore completely evaluated, including confluence, mixed argument derivatives, and anchored rectangular primitives. The divisor obstruction is an impossibility result for this certificate class. It is not an impossibility theorem for all summation identities involving an incompatible pair of denominators. Larger support, different auxiliary terms, and independently variable Hurwitz orders remain separate problems.

The Stieltjes question. Question 9 of the earlier *Polygamma Tail Identities* report excluded zero argument-derivative orders from its positive-order product generator [4]. Theorem 8.1 fills the edge with exactly one zero derivative order. Together with the inherited positive-positive formula, the uniform dictionary (8.10) covers every same-point quadratic product with positive total argument-derivative order. All spectral derivatives in that dictionary are evaluated at the single integer $w = r + t + 1$; ordinary zeta values at other positive integers occur in the coefficients. The double-undifferentiated germ $H_{1,1}$ and the nonzero periodic-translation formulas were already available in the unified manuscript [1]. Equation (8.23) checks compatibility with the former. The translated products and their contact corrections retain their original attribution.

The already incorporated Stieltjes source corrections, general digamma-power identities, Gamma normal forms, and convolution towers are not relabelled as new discoveries. The Gaussian S_6 and revised S_8 conjectures are also unaffected. Evaluating another conductor-four tail does not evaluate either of those separately defined periods.

11.2 Normalization and endpoint requirements for integration

Three conventions are essential when moving these results into the unified manuscript. First, the finite summation identities retain their upper endpoint until its limit is justified. In particular, (2.13) and (3.23) record information that can survive analytic continuation beyond the ordinary convergence region. A continued expression is not thereby the sum of the pointwise-specialized summands.

Second, meromorphic expressions must be combined before taking a removable value or an order derivative. The Mellin factors involving $s\zeta(s + 1)$ are entire combinations, including at $s = 0$. At the Stieltjes edge, both resonant local terms are subtracted jointly. Substituting a parameter equal to one into a formula proved only for larger integer orders does not prove the extension. Likewise, merging two separated singular points changes the local subtraction problem. It requires a collision theorem, rather than literal substitution into a translated finite-part formula.

Third, the finite part uses the original coordinate x and the constant term of the cutoff integral.

Integration by parts therefore has the explicit endpoint law (8.20). For example,

$$\mathrm{FP}\int_0^1 \{\gamma_1(x)\gamma_0'(x)+\gamma_0(x)\gamma_1'(x)\}\,dx = -\zeta(2)-\zeta'(2),$$

which is not zero. The Stieltjes convention (1.1) fixes every sign in this statement. Similarly, differentiating the weighted kernel $K(z,w)$ requires differentiating its factor $2n+z+w$ as well as both tails. The additional terms in (3.15) are part of the identity. These are requirements for correct reuse, not assertions that the source theorems violated them.

11.3 What the verification actually establishes

The order-ten certificate proves a nonzero retained determinant 60407 modulo 65537, and an independent reconstruction gives 15578 for the corresponding original determinant modulo the same prime. The order-twelve certificate covers all 11513 integers $7 \le b \le 11519$; its two independent implementations agree on every residue. The infinite remaining range is handled by the written threshold proof. These are exact arithmetic certificates. The bilinear/kernel checks include 144 rational cutoff identities, 72 rational kernel cutoff identities, and 3125 finite divisor strings, in addition to the symbolic cancellation and determinant checks. Its 21 numerical diagnostics at 50-digit precision have maximum absolute discrepancy below $5.4\cdot 10^{-51}$. The Mellin verifier records 2002 exact rational or algebraic checks and 11 direct integer-order or scaled quadratures. The Stieltjes verifier records 81 exact endpoint identities, four printed-formula checks, and five direct finite-part quadratures, with one repeated at a second splitting point. Its largest recorded absolute discrepancy is less than $1.3\cdot 10^{-46}$ at 55-digit working precision. The quadratures are numerical diagnostics, not interval certificates, and do not replace the holomorphy arguments for arbitrary complex orders or all-index derivatives. The supplied logs distinguish these two kinds of evidence. The combined `verify_all.py` run completed successfully.

11.4 Twelve concrete questions beyond the completed results

The following questions supplement the Mellin-specific programme already given above. They ask for structural identities, finite reductions, or complete nonvanishing arguments. No claim of global novelty or of arithmetic independence is built into their formulation.

R1. A uniform reconstruction mechanism. Find a determinant identity or a recurrence proving $\det M_{a,b} \neq 0$ for every $1 \le a < b$, or determine an exact exceptional set if one exists. The completed families through smaller order twelve constrain any proposed formula. At present the highest Bernoulli-degree determinant explains eventual nonvanishing for fixed a, but does not exclude cancellation by the lower degrees uniformly in both parameters. A mechanism preserving those lower-degree terms would resolve R2 of the quartic report, rather than merely enlarging the range of computations.

R2. The moving boundary in the gap-six family. Derive a bordered-Hankel or comparable exact formula for $M_{a,a+3}$ that retains all lower-index terms and the omitted-binomial conventions at the boundary. The fixed-order contraction used in the source does not justify omitting this information for arbitrarily large a at the fixed separation. A recurrence in a with explicit nonzero initial data would be particularly useful. This is a sharply defined intermediate problem between the existing gap-four theorem and unrestricted unequal orders.

R3. Closed reconstruction coefficients at orders ten and twelve. For $a = 5, 6$, replace an unevaluated matrix inverse in the reconstruction of a prescribed zeta coordinate by a uniform coefficient formula. The target is a fixed-size kernel or product expression for the polynomial weight, including the ordinary-integral subtraction constants. Such a formula would turn the existence theorem for each larger order into an immediately usable family of polygamma integrals. Producing another determinant ratio is only useful if its minors themselves admit a proved finite reduction.

R4. Minimal support beyond the adjacent bilinear class. Allow a finite collection of products $E_{n+i}F_{n+j}$ with rational coefficients and require the difference of the certificate to contain only a prescribed nonzero bilinear term. Determine the smallest support needed, modulo identities already implied by the tail recurrences. Does the divisor obstruction of Proposition 2.2 persist, or can an incompatible pair such as generic separately shifted pure powers be accommodated? A necessary-and-sufficient criterion would distinguish a genuine new summation mechanism from a rewriting of the two-product certificate.

R5. Finite ranks for longer adjacent denominator ladders. For

$$L_d(z) = \sum_{j\geq 0} \frac{(-1)^j}{(j+z)_d}, \qquad \Re z > 0, \quad d \geq 2,$$

determine the minimal separated rank of the weighted two-shift kernels obtained from admissible certificates. Even d admits a pairing of adjacent linear factors; odd d encounters the divisor obstruction at generic independent shifts. The rank-two formula for $d = 2$ suggests an exact finite determinant hierarchy. The objective is a minimal rank theorem and explicit confluent determinants for every admissible d, not just a separable expansion whose length has not been proved optimal.

R6. Three independently colored tails. Classify finite certificates built from neighboring products of three tails with independent denominator polynomials and colors. State which quadratic and linear residual tails must survive and give the complete finite endpoint. The previously developed powers of a single tail do not settle this independent-data problem. A first substantive target is a three-shift quadratic family, with a specified reduction to depth-two sums and one-factor functions; unrestricted depth-one closure should not be assumed.

R7. The continued alternating-square law at negative orders. Starting from its exact finite endpoint, construct a jointly holomorphic subtraction formula at each nonpositive tail order and derive every order derivative there. The already documented continued value $1/4$ at order zero, despite vanishing pointwise summands, is the compulsory normalization test [2]. The desired identities should identify the finite Euler-polynomial contributions and the spectral logarithmic terms separately. This would give exact negative-order identities without confusing continuation with an ordinary series.

R8. Independent nonperiodic Stieltjes shifts. Construct an order-generating formula for the ordinary integrals

$$\int_0^1 \gamma_m^{(r)}(x+a)\gamma_n^{(t)}(x+b)\,dx, \qquad \Re a, \Re b > 0,$$

with a and b independent. These are not the inherited periodic translations $\{x+a\}$. Begin with the antisymmetric first-derivative combination, since its symmetric companion is an endpoint derivative. Specify a finite function class or a genuinely new two-parameter master, and prove

how its diagonal and boundary limits recover the appropriate coincident finite parts. The independent shift difference is the missing variable in the classical coincident product kernel.

R9. Cubic Stieltjes products with independent indices. Determine an explicit holomorphic germ for FP $\int_0^1 \gamma_m^{(r)}\gamma_n^{(t)}\gamma_j^{(v)}\,dx$ when $r+t+v>0$. A first nontrivial target is FP $\int_0^1 \gamma_1(x)^2\gamma_0'(x)\,dx$. The all-index digamma-power results concern a different restriction of this problem. Fourier or Mellin analysis suggests that genuine multiple-zeta order derivatives may be needed. A successful theorem should identify that alphabet and all local counterterms before claiming a finite reduction; the single-spectral-argument quadratic closure does not imply its cubic analogue.

R10. Universal relations among the quadratic finite parts. For fixed total Stieltjes index and fixed positive total argument derivative, classify all relations implied by the exact coefficient dictionary. Determine whether exchange symmetry and the endpoint identity generate all of them, or whether further relations remain. The cancellation of the highest zeta derivative in (8.18) gives a concrete first test. Seek a uniform generating basis and dimension in a declared formal coefficient alphabet. This would explain cancellations systematically while avoiding an unproved claim about independence of evaluated constants.

R11. Simultaneous and iterated triple collisions. For three periodically translated Stieltjes factors, compare the constant term obtained by a simultaneous fixed-coordinate collision with the two possible successive pairwise collisions. Compute any difference as an explicit local distribution and test whether a canonical correction makes the resulting composition associative. The existing quadratic contact and dilation laws [1] are mandatory compatibility conditions, not a proof of this three-factor assertion. The question is meaningful only after the two-variable cutoff path and the retained finite term are specified; those choices are part of the proposed theorem.

R12. A finite evaluation of the logarithmic Gamma correction. The inherited function $S(a,b)=\sum_{n\ge0}\arctan^2(b/(n+a))$ already has an ordinary Gamma-integral representation and exact shift and distribution laws [2]. Find a proved finite evaluation at a nonzero mass, for example on a rational-shift locus with $e^{-2\pi b}$ algebraic, in an explicitly stated collection of polylogarithm values, order derivatives, and Gamma values. The existing small-mass series and integral remain valid answers at generic parameters. A finite special-value reduction would be an additional identity and would clarify which extra functions the higher spectral defects actually require.

11.5 Proposed repository integration

The accompanying integration note proposes separate inserts for the certificate classification, the shifted kernels and primitives, the Mellin resonances, the Stieltjes edge, and the two reconstruction families. It records their exact source dependencies and the status sentences that should change. The current manuscript should retain its established collision, convolution, and Gamma-normal-form chapters, with cross-references to the added edge theorem. The complete verification records should travel with the relevant inserts, including the finite-field residue list and its independent replay. No repository files are modified by this delivery.

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[1] ProveIt project, *Polylogarithms manuscript*, repository revision `da2792fc9cc8ef19c6a314be6c65e1437fbdc66b`, 11 October 2026. Pinned manuscript directory. The revision's 91 chapter files were indexed for the present audit.

[2] ProveIt project, *Alternating Tails and Spectral Defects*, incoming research report, 11 October 2026. In particular, the two-product research question and Theorem `rt:rank10` in `sections/07_rank_refinement.tex`. Pinned source archive.

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[4] ProveIt project, *Polygamma Tail Identities*, incoming research report, 11 October 2026. In particular, Theorem `hprod:stieltjes-products` in `sections/06_hurwitz_products.tex`, and paragraph 9 of `sections/09_audit_research.tex`. Archive at its containing commit. The archive's own baseline is different from that containing commit; see the supplied provenance record.

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[11] NIST Digital Library of Mathematical Functions, *§5.15, Polygamma Functions*. https://dlmf.nist.gov/5.15.

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